WildfireGuardian: Mathematical Methods

A step-by-step account of retrospective arrival reconstruction,
hourly transition prediction, flame residence, and expected radiant heat

Methods document grounded in the current project implementation

8 October 2026 (Asia/Seoul)

Abstract

This document specifies the maintained WildfireGuardian workflow as it exists on 8 October 2026. It follows the calculation from published satellite-based time-of-arrival reconstructions to aligned environmental tensors, two trained environment-conditioned spatial models, assumed active-flame probabilities, and an expected incident radiant-heat field. Each mathematical object is tied to its units, spatial support, time convention, validity conditions, and implementation. The nonnegative exponential-link model and signed direct-score model are treated separately. Derivations distinguish reconstructed labels, learned next-hour transition probabilities, deterministic residence scenarios, and the additional physical and timing assumptions required for radiation. Saved comparisons and demonstrations are reported as recorded evidence, not as fresh model fitting or independent physical validation. In particular, 375 m raster spacing and hourly output do not imply corresponding observation accuracy; a cumulative burned state is not an active-flame map; and a mean radiant flux does not determine a heat-exceedance probability.

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1 Scope, objects, and the calculation being performed

1.1 Exactly which workflow is described

The scope is the current root `README.md`, the maintained Python packages `wildfire_burned_area_hourly/` and `expected_heat_flux/`, their saved experiments, and the publisher reconstruction code retained under `wildfire_burned_area_hourly/sources/zenodo_21788344/extracted/code/`. The published reconstruction is an upstream method. The local bundle consumes its supplied rasters and constructs labels from them; ordinary model inference does not re-run kriging. Other historical projects, routing algorithms, and previous neural-network packages are outside this document's scope.

There are six successive operations:

1. Estimate an arrival time at each covered location from event-wide active-fire detections and a known final perimeter, using the published variance-weighted ordinary-kriging procedure.
2. Compare arrival time with each snapshot's event-relative clock to obtain cumulative burned-state and hourly transition labels.
3. Align environmental sources to the original rectangular grid, retaining each source's coverage, units, native support, and categorical vocabulary.
4. Fit or load an environment-conditioned spatial model for a currently unburned target cell's first new burning during the next hour.
5. Supply a flame or burning duration and an arrival-time assumption to convert that transition probability into activity at an instant or mean activity over an interval.
6. Supply dry fuel, combustion, radiation, and geometry assumptions to integrate the activity-weighted emitting sources into a radiant-heat field.

Every later operation depends on the earlier contracts, but it does not erase their uncertainty or turn retrospective information into live observations. The learned probability model receives only current-state and current environmental tensors at inference. Nevertheless, the current burned-state tensor itself was reconstructed using final extent and event-wide information. Restricting tensor timestamps to the cutoff does not make that input operationally available at the cutoff.

1.2 Global notation and units

The following notation is used throughout. Sections may introduce additional locally defined indices; the domain and units are stated where introduced.

Symbol	Definition and units
e	A fire event with its own raster, coordinate reference system, and time origin.
$\mathcal{G}_e$	The finite rectangular grid of event e; a cell index is $i = (r, c)$ with zero-based row r and column c.
S_i	The horizontal ground footprint of cell i, in a projected metric plane; its area is $\lvert S_i \rvert$ in m^2.
x_i	The projected horizontal center of S_i, in metres.
$t_{0,e}$	Event-specific UTC origin of the arrival raster.
t, $h_e(t)$	Absolute UTC snapshot time and elapsed hours $h_e(t) = (t - t_{0,e})/(3600\ \mathrm{s})$.
$\widehat{a}_i$	Reconstructed arrival time, in hours since $t_{0,e}$; defined only where its raster mask is valid.

Symbol	Definition and units
M_i^A	Boolean arrival-coverage indicator; one denotes a known reconstruction, zero denotes unavailable arrival.
$B_i(t)$	Reconstructed cumulative burned state, $\mathbf{1}\{\widehat{a}_i \le h_e(t)\}$ where $M_i^A = 1$; dimensionless.
$Y_i(t)$	Binary label for new burning during $(t, t + 1\text{ h}]$, on a known currently unburned target.
$E_i(t)$	Ordered environmental feature vector; raw continuous entries retain their recorded physical units and one-hot entries are dimensionless.
$s_i(t)$	Unclipped signed direct-model score, dimensionless; it may lie outside $[0, 1]$.
$\lambda_i(t)$	Nonnegative exponential-model rate in h^{-1}, under a frozen current input state.
$q_i(t)$	Usable next-hour new-burn probability, dimensionless in $[0, 1]$, or unavailable with a false prediction mask.
D	Duration of the heat forecast interval in seconds; the composed current model requires $D = 3600$ s.
T_i	Supplied positive conditional burning duration for the heat calculation, in seconds.
τ_i^{fl}	Surface-flame residence time in seconds; it equals T_i only when explicitly made consistent by the caller.
$p_i(u)$, $\overline{p}_i$	Active-burning probability at elapsed seconds u and its time average over $[0, D]$; dimensionless.
w_i, f_i, $H_{c,i}$	Dry fuel loading in $\mathrm{kg\,m^{-2}}$, consumed fraction in $[0, 1]$, and effective heat of combustion in $\mathrm{J\,kg^{-1}}$.
Q_i	Heat-release-rate density conditional on modeled activity, in $\mathrm{W\,m^{-2}}$.
χ_i	Radiative fraction of conditional heat release, dimensionless in $[0, 1]$.
$\mathcal{T}$	Atmospheric transmissivity, dimensionless in $[0, 1]$; distinct from duration and from flame residence.
$\Omega_i(x)$	Solid angle of elevated source footprint S_i seen from receiver point x, in steradians.
$F(x, u)$	Incident radiant flux at a horizontal point receiver, in $\mathrm{W\,m^{-2}}$; mean flux and endpoint flux are different outputs.
$\mathcal{E}(x)$	Time-integrated expected incident radiant energy, in $\mathrm{J\,m^{-2}}$.

The code field named `atmospheric_transmissivity` is denoted by $\mathcal{T}$ here even when a later equation or quoted method uses another letter. The field `burning_duration_s` is T_i; the residence object's `duration_s` is τ_i^{fl}. A notation reused within a section is explicitly identified. Tensor time length is denoted L, and batch size N_b, to avoid confusing these array dimensions with physical duration or burned state.

1.3 Six statements that are not interchangeable

1. A sensor detected active fire at a location and acquisition time.
2. A retrospective arrival reconstruction is valid at a raster cell.
3. A raster cell is classified as cumulatively burned at a snapshot time.
4. A model assigns probability q_i to its first new burning in the next hour.
5. A timing-and-residence scenario assigns probability $p_i(u)$ to a representative location being actively flaming at an instant.

6. A radiation scenario gives an expected incident flux at a receiver.

The numerical outputs support only the corresponding statement. For example, $B_i = 1$ does not imply present activity, and $q_i = 0.8$ does not imply an 80% chance of flames at the one-hour endpoint. A validity mask is required even when a stored numeric value is zero.

2 Step 1: published retrospective arrival-time reconstruction

2.1 Source product and observation preprocessing

The retained deposit supplies 30 Korean rasters (15 fires times arrival and kriging standard deviation) and 58 U.S. rasters (29 fires times the same two quantities). Korean output grids use EPSG:5179 and U.S. output grids use EPSG:5070. Both have 375 m output spacing. The source is the dataset and code identified by DOI 10.5281/zenodo.21788344 [1, 2]. This document explains the archived implementation; it does not claim that the raw detections or aerial reference perimeters were downloaded again or that the published study was independently reproduced.

Let an input detection record contain longitude, latitude, UTC acquisition time, and sensor. The archived `preprocess.select` removes fully identical rows, retains times within a closed collection interval $[t_{\min}, t_{\max}]$, and uses strict spatial inclusion in the final perimeter. Geostationary detections are selected inside a perimeter expanded by 2000 m; VIIRS uses no expansion. Boundary points are excluded by the implemented strict containment predicate.

U.S. analysis coordinates are EPSG:5070 coordinates divided by 1000, in km. Korean analysis coordinates are the source's approximation

$$x = 111\,\ell\cos(\varphi_0), \qquad y = 111\,\varphi, \tag{1}$$

where longitude ℓ, latitude φ, and event-center latitude φ_0 are in degrees and the outputs are in km. The cosine argument is converted to radians in code. These Korean *analysis* coordinates differ from the projected EPSG:5179 coordinates of the delivered raster.

VIIRS detections are grouped using

$$(g_x, g_y) = (\mathrm{round}(x/0.375), \mathrm{round}(y/0.375))\,. \tag{2}$$

The implementation uses NumPy rounding, including its ties-to-even convention. Each occupied analysis cell contributes the median detection time; its representative location is the earliest detection's location, breaking time ties by smaller latitude and then longitude. Thus the temporal statistic is the *median*, whereas the representative spatial point comes from the *earliest* detection. Neither operation establishes exact physical ignition time.

2.2 Event-relative clocks

Each detection time is converted to hours after its event origin. In the U.S., the origin is midnight UTC on the day containing the start of the collection window, whose start is the first NIROPS survey minus 24 hours. For Korea, the origin is the Korea Forest Service situation-map ignition time. The authoritative local origin comes from GeoTIFF tags and event metadata. Never compare arrival numbers from two events before adding their respective origins. Valid U.S. arrival predictions may be negative: validity is read from NoData and coverage metadata, not inferred from the sign.

2.3 Empirical semivariogram and fitted covariance

Let z_j be the detection-cell summary time in hours at analysis point $\xi_j \in \mathbb{R}^2$ in km, for $j = 1, \ldots, n$. For each nonempty lag bin $\mathcal{P}_k$ of unordered point pairs, define

$$\widehat{h}_k = \frac{1}{|\mathcal{P}_k|} \sum_{(j,l)\in\mathcal{P}_k} \|\xi_j - \xi_l\|_2, \qquad \widehat{\gamma}_k = \frac{1}{2|\mathcal{P}_k|} \sum_{(j,l)\in\mathcal{P}_k} (z_j - z_l)^2. \tag{3}$$

The code uses 20 equal-width bins between the smallest and largest pair distance, with a final edge extended by 0.001 km. Empty bins are omitted. For more than 15,000 observations it accumulates the same pair statistics in chunks instead of materializing all pairs at once.

The fitted model is

$$\gamma(h) = p\left(1 - e^{-3h/r_0}\right) + c_0, \qquad p \geq 0, \quad r_0 > 0, \quad c_0 \geq 0, \tag{4}$$

where p is the partial sill in h^2, c_0 is the nugget in h^2, and r_0 is the effective range in km. At $h = r_0$ the structured term is $(1 - e^{-3})p \simeq 0.9502p$. Bounded SciPy least squares uses the `soft_l1` loss on residuals $\gamma(\widehat{h}_k) - \widehat{\gamma}_k$ with its default unit scale, tight termination tolerances 10^{-15}, and at most 2,000,000 function evaluations. The initial values are

$$\left(\max_k \widehat{\gamma}_k - \min_k \widehat{\gamma}_k, \tfrac{1}{4}\max_k \widehat{h}_k, \min_k \widehat{\gamma}_k\right), \tag{5}$$

and upper parameter bounds are $(10\max_k \widehat{\gamma}_k, \max_k \widehat{h}_k, \max_k \widehat{\gamma}_k)$. The returned nugget is floored to $\max(c_0, 0.001p)$. A failed convergence status raises an error. The implemented covariance is

$$C(h) = \begin{cases} p + c_0, & h = 0, \\ pe^{-3h/r_0}, & h > 0. \end{cases} \tag{6}$$

The nugget at zero is an explicit implementation convention, not a claim that the semivariogram expression at zero directly equals the variance of a difference between a variable and itself.

2.4 Local ordinary kriging with observation-dependent error variance

At a target point ξ, choose the nearest $m = \min(50, n_{\text{all}})$ observations from the combined VIIRS and geostationary table. At least three neighbors are required for a finite prediction. The production procedure also requires at least five VIIRS summary cells to fit an event product. The variogram is fitted on VIIRS summary cells, before adding geostationary points to the kriging system.

Let v_j denote assigned error variance, in h^2:

$$v_j = \begin{cases} 0, & \text{VIIRS summary cell}, \\ 27.4\max(c_0, 1\ \mathrm{h}^2), & \text{geostationary point}. \end{cases} \tag{7}$$

The source's factor 27.4 is a rounded resolution-based inflation factor; its comment motivates it by $(2/0.375)^2 - 1$. It is a prescribed error model, not an empirically calibrated uncertainty for every geostationary observation. Write $\epsilon = 10^{-4}p$ for the numerical ridge, and set

$$K_{jl} = C(\|\xi_j - \xi_l\|_2) + \mathbf{1}\{j = l\}(v_j + \epsilon), \qquad c_j = C(\|\xi_j - \xi\|_2). \tag{8}$$

The weights $\alpha \in \mathbb{R}^m$ and Lagrange multiplier $\mu \in \mathbb{R}$ solve

$$\begin{pmatrix} K & \mathbf{1} \\ \mathbf{1}^\top & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \mu \end{pmatrix} = \begin{pmatrix} c \\ 1 \end{pmatrix}. \tag{9}$$

The last row enforces $\mathbf{1}^\top\alpha = 1$, the ordinary-kriging unbiasedness constraint under an unknown constant mean. To derive the system, minimize the mean-squared error

$$V(\alpha) = C(0) - 2\alpha^\top c + \alpha^\top K\alpha \tag{10}$$

subject to that constraint. Differentiating the Lagrangian $V(\alpha) + 2\mu(\mathbf{1}^\top\alpha - 1)$ yields $K\alpha + \mu\mathbf{1} = c$, exactly the first block row. Therefore

$$\widehat{a}(\xi) = \alpha^\top z, \qquad \widehat{V}(\xi) = \max\{p + c_0 - \alpha^\top c - \mu, 0\}. \tag{11}$$

The variance simplification follows by multiplying the first block row by $\alpha^\top$: $\alpha^\top K\alpha = \alpha^\top c - \mu$. The nonnegative floor is a numerical safeguard. The delivered uncertainty raster is $\sqrt{\widehat{V}}$ in hours. It is a model-based kriging standard deviation; the local downstream methods do not treat it as a calibrated Gaussian posterior, sample arrival-time realizations, or convert it into flame-confidence intervals.

2.5 Reference evaluation and internal cross-validation

The upstream code compares five interpolation configurations: VIIRS-only IDW, VIIRS-plus-geostationary IDW, VIIRS-only ordinary kriging, pooled unweighted ordinary kriging, and the variance-weighted pooled method above. IDW uses 20 nearest observations and weights proportional to $\max(\|\xi_j - \xi\|_2, 10^{-10}\ \text{km})^{-2}$. These are reconstruction comparators, distinct from the later learned exponential and signed direct probability models.

U.S. external reference arrival is the time of the first aerial infrared survey whose cumulative union encloses a cell center. It is stepwise at survey times, not continuously observed physical arrival. At survey time t_s, let H_s, M_s, and F_s be counts of intersection, reference-only, and prediction-only burned cells, respectively, within the final reference support. The implemented Sørensen coefficient is

$$\mathrm{SC}_s = \frac{2H_s}{2H_s + M_s + F_s}. \tag{12}$$

The default growth window retains surveys with reference count $\max(1, 0.01N) \le N_s < 0.90N$, where N is the final reference count. An event score is the median over eligible surveys, rounded to four decimals.

Spatial-block internal validation withholds VIIRS cells while retaining geostationary observations. Block width is

$$b = \min\{30, \max(5, r_0/2)\} \quad \text{km}. \tag{13}$$

The U.S. procedure withholds each occupied block, requires at least five training cells, refits the variogram for each fold, and uses a fold only when all five configurations return finite predictions. The Korean procedure uses at most ten grouped folds, at least 20 training and three validation cells per fold, and adoption conditions including whole-data range at least 0.5 km, three occupied blocks, ten pooled validation pairs, and nonzero observation and prediction variance. Withheld evaluation is an unavailable assessment, not proof of inaccurate reconstruction.

For residuals $d_j = \widehat{z}_j - z_j$, timing scores are

$$\mathrm{ME} = \frac{1}{n}\sum_j d_j, \qquad \mathrm{MAE} = \frac{1}{n}\sum_j |d_j|, \tag{14}$$

in hours, with Pearson correlation also reported. Across events, paired exact two-sided sign tests omit zero differences, and Holm adjustment controls the family of ten configuration pairs. If ordered raw values are $p_{(1)} \le \cdots \le p_{(10)}$, adjusted values are $\widetilde{p}_{(k)} = \max_{j \le k} \min\{1, (11-j)p_{(j)}\}$. Percentile intervals for median paired differences and Spearman correlations resample entire event pairs 10,000 times with saved seed 20260906. These source-study statistical procedures were read from retained code; they were not rerun to author this document.

3 Step 2: converting arrival time to hourly labels

3.1 Cumulative state and transition intervals

For $M_i^A = 1$, the reconstructed cumulative state is

$$B_i(t) = \mathbf{1}\{\widehat{a}_i \le h_e(t)\}. \tag{15}$$

It is nondecreasing in t. At the precise reconstructed arrival time it is already one. Where $M_i^A = 0$, the physical state remains unknown even though a storage tensor may contain zero. The loader's field `newly_burned_area` at snapshot t means the *preceding* hour:

$$N_i^{\text{past}}(t) = \mathbf{1}\{h_e(t) - 1 < \widehat{a}_i \le h_e(t)\}. \tag{16}$$

The model's training label instead means the *following* hour:

$$Y_i(t) = \mathbf{1}\{h_e(t) < \widehat{a}_i \le h_e(t) + 1\} = B_i(t + 1\text{ h}) - B_i(t). \tag{17}$$

This equality assumes the same valid reconstructed arrival is used at both times. Training candidates have known $B_i(t) = 0$, so the future burned value also equals $Y_i(t)$ on those candidates. The training preparation loads current environmental and burned-state contexts in 24-hour chunks. For a chosen cutoff, it computes the next-hour target state directly from the arrival raster using the next elapsed-hour boundary, then checks equality with the independently enumerated candidate label. It does not use the loader's preceding-hour field as the forecast target. The precise eligibility and sampling procedure is given in the probability-method section.

3.2 Mask encoding, area, and temporal precision

Exported categorical hourly masks use code 1 for estimated burned, code 0 for estimated not yet burned *inside* valid reconstruction support, and code 255 for unavailable. Code 0 outside a tensor's validity mask cannot be interpreted as measured unburned ground. At square 375 m spacing,

$$|S_i| = 375^2 = 140625\text{ m}^2 = 14.0625\text{ ha}. \tag{18}$$

The cumulative raster-support area is $\mathcal{A}_e(t) = 14.0625 \sum_{i:M_i^A=1} B_i(t)$ hectares. It counts entire projected supports selected by a center-in-perimeter rule. It need not equal the exact vector-perimeter area or administrative forest-damage area, and no such area equality or rescaling is imposed.

One-hour output spacing is a thresholding schedule for a continuous reconstruction, not a one-hour accuracy statement. For example, the retained Uljin regional metadata reports block-validation MAE about 10.44 hours and kriging standard deviation about 4.40–52.96 hours. The current active-flame example has much larger arrival uncertainty than its seconds-long residence scenario. Thresholding uncertainty at a finer clock does not remove it.

4 Step 3: aligning and loading environmental inputs

4.1 Spatial alignment and the original rectangular grid

The environmental package retains each event's original arrival-raster shape, CRS, and affine transform. The environment occupies the *rectangular* grid and is not cropped to future final burned extent. Arrival labels retain their own reconstruction mask. Write the affine coefficients as (a, b, c, d, e, f), using the conventional raster corner transform. For zero-based row r and column c_{col}, the cell center is

$$\begin{pmatrix} x_{r,c_{\text{col}}} \\ y_{r,c_{\text{col}}} \end{pmatrix} = \begin{pmatrix} a & b \\ d & e \end{pmatrix} \begin{pmatrix} c_{\text{col}} + 1/2 \\ r + 1/2 \end{pmatrix} + \begin{pmatrix} c \\ f \end{pmatrix}. \tag{19}$$

The corner coordinate c in this equation is an affine coefficient, not the column index. The cell area is $|ae - bd|$ in square metres. On an extracted patch beginning at row r_0, column c_0, its affine transform is the event transform composed with pixel translation (c_0, r_0). Coordinates and masks are supplied along with values; padded spatial coordinates are NaN.

4.2 Source support and resampling

Input family	Source/support in this bundle	Alignment and interpretation
Weather	Archived IFS model values, roughly 9 km and hourly	Nearest returned native model-node center in the event metric plane; all its assigned pixels share the same node value.
Mixing depth	ERA5, about 0.25° and hourly	Nearest native-node mapping; boundary-layer height is a mixing-depth proxy.
Terrain	Copernicus GLO-30 DSM, nominal 30 m	Bilinear reprojection to 31.25 m intermediate grid, terrain derivatives there, then finite-value aggregation to 375 m.
Land cover	Copernicus global land-cover 2019, 100 m	Modal class for categorical layers; valid-pixel averages for continuous percent covers.
Canopy height	GLAD GEDI–Landsat 2019 map, about 30 m	Valid-pixel averages; map epoch and model limitations are retained.
Fuel classes/structure	U.S. LANDFIRE 2016 Remap, 30 m	Modal class for FBFM40/FCCS; averages for numeric encodings, followed by stated physical-unit conversions.
Reference dry loading	LANDFIRE FCCS lookup library	Lookup on the original native 30 m codes *before* valid-pixel averaging; coverage fraction stored separately.
Fuel moisture	U.S. gridMET 100-hour and 1000-hour dead-fuel moisture, about 4 km, daily	Nearest spatial resampling; values are percent of dry fuel mass. These are not live/fine fuel moisture or hourly measurements.
Biome	RESOLVE Ecoregions 2017 polygons	Static broad-scale biome assignment at cell centers; fixed 14-class one-hot vocabulary for trained inputs.

Resampling coarser sources to 375 m does not produce independent 375 m weather observations. Static source years may precede or follow an event; the data do not establish event-date measured fuel or vegetation. Some environmental variables are available to the general loader but are not inputs to the saved 19-channel predictor. In particular, a moisture raster in the bundle is not evidence of a learned moisture effect in that fitted model.

For a target center x_i and returned native-node centers η_j, weather mapping is

$$j(i) = \operatorname{argmin}_j \|x_i - \eta_j\|_2, \qquad E_i(t) = V_{j(i)}(t). \tag{20}$$

The saved node-index and node-distance rasters retain the mapping. Selection uses the metric projection, not naive Euclidean distance in latitude/longitude. For continuous native rasters, a valid-support average can be expressed as

$$\overline{v}_i = \frac{\sum_j \omega_{ij} m_j v_j}{\sum_j \omega_{ij} m_j}, \tag{21}$$

where $\omega_{ij} \geq 0$ are reprojection overlap weights and m_j is native validity. An empty denominator produces unavailable output. A modal class is the code with greatest aggregated valid weight, not the average of numeric class codes.

4.3 Derived meteorological and terrain features

If wind speed v is in $\mathrm{m\,s^{-1}}$ and meteorological direction θ is the direction *from which* wind blows, clockwise from north, the saved eastward and northward components are

$$u = -v \sin\theta, \qquad v_N = -v \cos\theta. \tag{22}$$

Angles are converted from degrees to radians. The negative signs convert "from" into vector direction "toward." For hourly preceding-interval precipitation P_k in mm, trailing precipitation is

$$P_k^{(m)} = \sum_{j=0}^{m-1} P_{k-j}, \qquad m \in \{24, 168, 720\}, \tag{23}$$

with all m values required. The dry-hour recurrence resets to zero when $P_k \geq 0.1$ mm and otherwise increments the previous count for finite precipitation. It is a lower bound when a dry spell reaches the beginning of downloaded history; missing source precipitation keeps the streak unavailable.

On the 31.25 m intermediate terrain grid, let z be elevation and g_x, g_s its finite-difference gradients in eastward and increasing-row (southward) directions. The source code computes

$$\alpha = \frac{180}{\pi} \arctan\sqrt{g_x^2 + g_s^2}, \qquad \beta = \operatorname{atan2}(-g_x, g_s) \pmod{2\pi}. \tag{24}$$

α is slope in degrees; β is downslope aspect clockwise from projected-grid north. Near-flat fine pixels with gradient magnitude below 10^{-5} have unavailable aspect. Coarse circular averaging uses

$$e_i = \overline{\sin\beta}, \quad n_i = \overline{\cos\beta}, \quad \beta_i = \operatorname{atan2}(e_i, n_i), \quad \rho_i = \sqrt{e_i^2 + n_i^2}, \tag{25}$$

rather than an arithmetic mean of angles; $\rho_i < 10^{-5}$ makes coarse aspect unavailable. Terrain position index is $z_i - \overline{z}_{\mathcal{N}_i}$, and neighborhood elevation standard deviation is $\sqrt{\max(\overline{z^2}_{\mathcal{N}_i} - \overline{z}^2_{\mathcal{N}_i}, 0)}$, using available cells of a 9×9 neighborhood, nominal width 3375 m. Vegetated cover is the sum of tree, shrub, grass, and crop percentages with all four required. Its 3×3 neighborhood mean is a coverage proxy, not a measurement of subpixel fuel connectivity.

4.4 Temporal support and the completed-day convention

Snapshot clocks and dynamic feature supports are distinct. Instantaneous hourly fields are tagged $[t, t]$. Precipitation, rain, radiation, and evapotranspiration fields summarize the preceding hour $[t - 3600, t]$. Trailing rainfall support begins 24 hours, seven days, or 30 days before t. The loader stores the support start and end as Unix seconds for each channel. These metadata describe source support, not verified publication availability.

Daily gridMET labels correspond to nominal midnight-to-midnight Mountain Standard Time, with a fixed 07:00 UTC boundary. Let s_d be the UTC start of daily source interval d. The default `completed_day` policy chooses the latest d with

$$s_d + 86400 \leq t < s_d + 2 \cdot 86400. \tag{26}$$

No missing date is extrapolated, and no selection is extended beyond the downloaded series. The optional `containing_day` policy instead chooses the interval with $s_d \leq t < s_d + 86400$; a full-day value then includes times after a within-day cutoff and is suitable only under an explicit retrospective interpretation. Repeating a selected daily or static value over a sequence does not turn it into a newly observed hourly quantity.

4.5 Categorical encoding and independent validity masks

For categorical source Z, freeze ascending distinct class codes $\mathcal{C}_Z = (c_1, \ldots, c_J)$. At a known source value z_i, indicator channel j is

$$X_{i,j} = \mathbf{1}\{z_i = c_j\}. \tag{27}$$

Known membership implies exactly one active indicator, with every indicator's validity mask true. Missing, absent, unknown-category, or padded membership gives all-zero indicators and false masks for the entire group. A valid numeric class code zero is preserved when it belongs to the vocabulary; numeric zeros are not automatically missing.

Bundle-wide discovery before event/time filtering prevents a changing schema between subsets. The full recorded loader schema expands 72 source fields into 741 channels: 65 continuous channels and 676 indicators. This is a schema width, not a statement that all channels are available for every event. The fitted predictor freezes a different subset: five continuous channels and 14 biome indicators, for 19 inputs. The checkpoint saves their order, categories, groups, and normalization. Re-discovering a smaller vocabulary at inference would change parameter meaning and is rejected.

The quality codes are 0 valid, 1 out of range, 2 source missing, 3 channel absent, 4 spatial padding, 5 temporal padding, and 6 unknown category. Flagged raw observations remain in source records but have false loader masks. A finite fill value is a storage convenience; its number carries no missingness semantics. The general loader returns environmental values and masks with shape $[L, C, H, W]$, current labels and masks with shape $[L, 1, H, W]$, and a collated batch adds a leading dimension N_b. Coordinates have shape $[2, H, W]$, ordered projected x then y. The `pixel_valid` mask describes padding, and `time_valid` describes padded times; neither is a substitute for environmental or arrival coverage.

4.6 Geographic bitmap and array exports

The image bundle contains 1,380 BMP files with geographic sidecars. Categorical meaning is in palette indices, not displayed RGB brightness. Exact continuous values are retained separately in `.values.npy`; quantized image colors must not be used to recover scientific precision. A world file lists A, D, B, E, C, F with upper-left *pixel-center* coordinates, giving

$$x = A\,c + B\,r + C, \qquad y = D\,c + E\,r + F. \tag{28}$$

This differs from the pixel-corner affine transform, whose center evaluation requires the half-pixel shift. CRS, coordinates, timestamps, range, and mask metadata must travel with a numeric export.

5 Step 4: exact next-hour transition targets and the statistical population

The probability experiments operate on the reconstructed burned-area bundle. Consequently, their labels describe transitions in a retrospective arrival-time reconstruction. They are not independently observed hourly ground-truth measurements. The code implementing this section is `wildfire_burned_area_hourly/probability_model/prepare_training.py`, `wildfire_burned_area_hourly/probability_model/filter_contact.py`, and `wildfire_burned_area_hourly/probability_model/cache.py`.[3]

5.1 Define each example before defining a model

Let e denote a fire event, $i = (r, c)$ a raster-cell index, and t an hourly UTC cutoff. In the global notation of Section 1.2, $A_e(i) = \widehat{a}_i$ and $a_e(t) = h_e(t)$ below. Let $A_e(i)$ be the

reconstructed arrival time at cell i, measured in hours from that event's recorded origin $t_{0,e}$, and let $a_e(t) = (t - t_{0,e})/(3600\text{ s})$ be the elapsed time at cutoff t. At cells with valid reconstructed arrival data, define the cumulative state

$$B_{e,t}(i) = \mathbf{1}\{A_e(i) \le a_e(t)\} \in \{0, 1\}. \tag{29}$$

This state remains one after arrival; it does not identify an actively flaming cell. A currently unburned cell has $B_{e,t}(i) = 0$. Its exact next-hour target is

$$\begin{aligned} Y_{e,t}(i) &= \mathbf{1}\{B_{e,t}(i) = 0,\ B_{e,t+3600\text{s}}(i) = 1\} \\ &= \mathbf{1}\{a_e(t) < A_e(i) \le a_e(t) + 1\}. \end{aligned} \tag{30}$$

The second expression applies only where arrival time is valid. Thus a target is positive precisely when arrival lies in the half-open interval $(t, t + 1\text{ hour}]$, interpreted through the event origin. A cell burned exactly at t is already burned and is excluded from the at-risk training population; a cell burned exactly at $t + 1$ hour gives a positive target.

The preparation code enumerates actual hourly timestamps from the catalog start through end. It uses `searchsorted(..., side="left")` to find the first hourly state satisfying $A_e(i) \le a_e(t)$. It checks that the current snapshot is exactly at the recorded cutoff, the next snapshot is exactly 3,600 seconds later and inside the event interval, and every dynamic input has an interval-end timestamp no later than the current cutoff. Fractional event origins are retained in the elapsed-time calculation rather than rounded to an integer hour.

5.2 Spatial support, masks, and contact

For an odd side length $K \ge 3$, put $R_K = (K - 1)/2$ and define the offset set

$$\mathcal{D}_K = \{-R_K, \dots, R_K\}^2, \qquad \mathcal{D}_K^* = \mathcal{D}_K \setminus \{(0, 0)\}. \tag{31}$$

Offsets are enumerated in row order. The source associated with target i and offset δ is $i + \delta$. This is the PyTorch cross-correlation convention; no reversal of the offsets is made. Define the number of currently burned neighboring cells by

$$n_{e,t,K}(i) = \sum_{\delta \in \mathcal{D}_K^*} B_{e,t}(i + \delta), \qquad C_{e,t,K}(i) = \mathbf{1}\{n_{e,t,K}(i) > 0\}. \tag{32}$$

Here $C = 1$ means contact with a previously burned cell within the support; it does not imply physical flame contact or an observed heat-transfer pathway.

Let $M^B_{e,t}(j)$ indicate that burned state is known at source cell j. Let $X_{e,t,c}(j)$ be environmental channel c at that cutoff, and let $M^X_{e,t,c}(j)$ indicate that its value is valid and finite. Before contact filtering, a training target must satisfy all of the following conditions:

1. Its current and exact next-hour arrival-derived states are known, its current state is zero, and the next hour is inside the catalog interval.
2. Every burned-state source in $i + \mathcal{D}_K$ is known.
3. Every selected environmental channel at every position in $i + \mathcal{D}_K$ is known and finite.
4. The entire square support lies inside the raster. The saved fitting contract uses unknown exterior boundaries and complete interior patches.

Let $\mathcal{U}_K$ denote the finite population of such cell-hours. The maintained contact-only experiments use the restricted population

$$\mathcal{U}_K^+ = \{(e, t, i) \in \mathcal{U}_K : C_{e,t,K}(i) = 1\}. \tag{33}$$

The empirical probability target is therefore conditional on complete reconstruction context and, in the maintained fits, on current burned-neighbor contact. Moreover, the arrival bundle supplies a final-perimeter-constrained retrospective support. Evaluation on this population does not estimate the probability of ignition anywhere in a surrounding landscape.

5.3 Why the contact restriction changes the task

Both architectures below sum coefficients multiplied by the current burned state, with no intercept after that sum. Therefore

$$n_{e,t,K}(i) = 0 \quad \Longrightarrow \quad \lambda_{e,t}(i) = 0, \quad q_{e,t}(i) = 0, \quad s_{e,t}(i) = 0. \tag{34}$$

This is an algebraic property, independent of training. In the earlier five-channel unrestricted 3×3 experiment, 5,549 of 20,345 positive test targets had no current burned neighbor: 27.2745% of positive targets. All of those targets necessarily received probability zero, so the exact Bernoulli negative log likelihood was infinite. The source for these counts is `probability_model/training_runs/20261007_1h_all44/fitted/metrics.json` relative to the burned-area package directory. Filtering to $\mathcal{U}_K^+$ removes this structurally unsupported part of the task. It does not repair the model's ability to predict disconnected arrivals; it defines a narrower task and must be reported as such.

6 Sample preparation, inverse inclusion weights, and fixed splits

6.1 Retain positives and sample negative cell-hours

Preparation first identifies candidate currently unburned cell-hours with complete arrival-derived burned context. It retains every positive candidate. Within each fire, negative candidates are divided into two strata: current contact and current no-contact. For a stratum h with N_h negative cell-hours, the code draws

$$m_h = \min(N_h, L_h) \tag{35}$$

uniformly without replacement. The recorded limits are $L_h = 5{,}000$ for contact negatives and $L_h = 2{,}000$ for no-contact negatives, with seed 7. Each selected negative has inclusion probability $\pi_u = m_h/N_h$ and weight $w_u = 1/\pi_u = N_h/m_h$. Every positive has $\pi_u = 1$ and $w_u = 1$. There is no draw when $N_h = 0$. The implementation samples integer time segments per cell using cumulative segment lengths; this is equivalent to uniform sampling from the stratum's disjoint cell-hour identities.

Environmental validity is checked after candidate sampling. Invalid selected examples are removed without replacement draws and without changing their weights. Contact filtering likewise retains the original weights exactly. For a fixed finite valid target population $\mathcal{V}$ and any fixed quantity f_u, let I_u indicate initial sampling. Then

$$\mathbb{E}_{\text{sampling}}\left[\sum_{u \in \mathcal{V}} I_u w_u f_u\right] = \sum_{u \in \mathcal{V}} f_u, \tag{36}$$

because $\mathbb{E}[I_u] = \pi_u$ and $w_u \pi_u = 1$. Filtering a deterministic valid subset preserves this identity for totals; a replacement draw would change the inclusion design and require a new calculation.

Most reported means use the ratio

$$\widehat{\mu}_f = \frac{\sum_{u \in \mathcal{S}} w_u f_u}{\sum_{u \in \mathcal{S}} w_u}, \tag{37}$$

where $\mathcal{S}$ is the retained sampled population. This is an inverse-inclusion-weighted ratio estimate, not an exhaustive full-grid mean. Equation (36) makes the numerator and denominator unbiased estimators of their respective totals for fixed f; it does not make their ratio exactly unbiased. It also does not establish independence between neighboring cell-hours or eliminate the effect of model fitting on the same training sample. The weighted rank and reliability diagnostics below are also sample-based descriptive estimates, with no reported confidence intervals.

6.2 Whole-season partitions

The split function is fixed by country and year before fitting:

Partition	Country-year seasons	Catalog events
Training	Korea 2022; United States 2020 and 2021	27
Validation	Korea 2023; United States 2022	7
Test	Korea 2025; United States 2024	10

All cell-hours of a fire remain in its assigned partition. One model per kernel pools the two countries; the maintained runs do not fit separate country-specific models. A catalog fire can contribute no eligible targets after complete-context and contact restrictions, so catalog event counts and contributing event counts differ.

For the 19-channel maintained native 3×3 cache, the saved counts are:

Partition	Contributing fires	Samples	Positives	$\sum w$
Training	25	205,841	99,850	3,867,244
Validation	5	28,318	12,936	745,565
Test	10	53,292	14,796	315,920

The test sample positive fraction is $14{,}796/53{,}292 = 0.277640$, whereas the weighted test prevalence is $14{,}796/315{,}920 = 0.0468346$. The large difference follows from retaining all positive targets while sampling negatives. Treating the sample fraction as the target-population prevalence would be incorrect. Counts are from `probability_model/training_runs/20261007_1h_biome_onehot_pooled/prepared/metadata.json`.

7 Input encoding and training-only normalization

7.1 The fitted probability models use nineteen channels

The probability-model input schema selects five continuous source fields: temperature at 2 m, eastward wind at 10 m, northward wind at 10 m, terrain slope in degrees, and tree-cover percentage. It also selects `biome.biome_class`. The frozen biome vocabulary contains class codes $1, \dots, 14$, expanded into fourteen distinct indicator channels:

$$X_{e,t,5+\ell}(j) = \mathbf{1}\{\mathrm{biome}_e(j) = \ell\}, \qquad \ell = 1, \dots, 14. \tag{38}$$

For a valid categorical cell, every indicator is exactly zero or one and their sum is exactly one. A missing biome is invalid; it is not represented by an all-zero known vector. Vocabulary and ordered channel names are frozen from the source legend and carried in cache metadata and the checkpoint. The integers assigned to biome classes are not treated as continuous ordered measurements. Thus $C = 5 + 14 = 19$ for these fits, even though the general loader can expose a broader environmental schema.

7.2 Weighted moments across training patches

Index the retained training examples by $u \in \mathcal{S}_{\mathrm{tr}}$. Write $X_{u,c,\delta}$ for channel c at offset δ in example u. Put $W_{\mathrm{tr}} = \sum_{u \in \mathcal{S}_{\mathrm{tr}}} w_u$. The code calculates

$$\mu_c = \frac{1}{W_{\mathrm{tr}} K^2} \sum_{u \in \mathcal{S}_{\mathrm{tr}}} w_u \sum_{\delta \in \mathcal{D}_K} X_{u,c,\delta}, \tag{39}$$

$$v_c = \frac{1}{W_{\mathrm{tr}} K^2} \sum_{u \in \mathcal{S}_{\mathrm{tr}}} w_u \sum_{\delta \in \mathcal{D}_K} (X_{u,c,\delta} - \mu_c)^2, \tag{40}$$

$$\sigma_c = \begin{cases} \sqrt{v_c}, & \sqrt{v_c} \geq 10^{-6}, \\ 1, & \sqrt{v_c} < 10^{-6}, \end{cases} \qquad Z_{u,c,\delta} = \frac{X_{u,c,\delta} - \mu_c}{\sigma_c}. \tag{41}$$

All moments use training examples only. Each target's weight applies equally to all K^2 positions in its patch. These are moments of the weighted patch distribution, not moments of unique raster cells: overlapping patches can contain the same physical cell multiple times. The calculation uses float64 accumulation and a second centered pass for variance, then saves float32 means and scales. One-hot channels receive the same normalization rule after their raw indicator validity is checked.

Validation, test, demonstration, and inference reuse the saved μ_c and σ_c without refitting. Each kernel has its own native training population and hence its own saved moments. Center-cropping a 9×9 evaluation patch to a smaller support does not change that smaller model's normalization parameters. This distinction matters in the common comparison.

8 Environmental kernel generator and the two output formulas

8.1 Shared position-dependent coefficient generator

The generator receives the normalized environmental $K \times K$ context of each target. With $d_h = 16$ hidden channels, define first-layer weights $U_{h,c,\delta} \in \mathbb{R}$, hidden biases $b_h \in \mathbb{R}$, second-layer weights $V_{\eta,h} \in \mathbb{R}$, and output biases $d_\eta \in \mathbb{R}$, where $h = 1, \ldots, d_h$, $c = 1, \ldots, C$, and $\delta, \eta \in \mathcal{D}_K$. At a target $u = (e, t, i)$ the exact center computation is

$$a_{u,h} = b_h + \sum_{c=1}^{C} \sum_{\delta \in \mathcal{D}_K} U_{h,c,\delta} Z_{u,c,\delta}, \tag{42}$$

$$H_{u,h} = \mathrm{SiLU}(a_{u,h}) = a_{u,h} \,\mathrm{sigmoid}(a_{u,h}), \tag{43}$$

$$z_{u,\eta} = d_\eta + \sum_{h=1}^{d_h} V_{\eta,h} H_{u,h}. \tag{44}$$

Here $\mathrm{sigmoid}(a) = 1/(1 + e^{-a})$. The quantity d_h is hidden width, distinct from the raster's row count H. The first layer is a $K \times K$ convolution and the second a 1×1 convolution with K^2 outputs. Applying these convolutions across a full grid produces a different vector $(z_{u,\eta})_\eta$ at every target. The same learned parameter arrays are shared across targets and events. The generator reads environmental data; the generated coefficients act on the burned-state map. The burned map is not an additional input channel to the generator. Leading time and batch dimensions are independent snapshots; these modules have no temporal recurrence.

Let $m_\eta = \mathbf{1}\{\eta \neq (0, 0)\}$. Both models fix their center coefficient to zero by multiplication by m_η. The output biases d_η are biases of the coefficient generator; they are not an additive intercept after the burned-source sum. No normalization makes neighboring coefficients sum to one.

The allocated trainable parameter count is

$$P_K = d_h C K^2 + d_h + d_h K^2 + K^2 = K^2\{d_h(C + 1) + 1\} + d_h. \tag{45}$$

For $C = 19$ and $d_h = 16$, P_K is 2,905, 8,041, 15,745, and 26,017 for $K = 3, 5, 7, 9$, respectively. The allocated center-output parameters are mathematically inactive because of the fixed mask. Holding d_h constant therefore does not hold model capacity constant across kernel sizes.

8.2 Original nonnegative rate and exponential-link model

The original architecture maps logits to nonnegative hourly contributions:

$$k_{u,\eta} = m_\eta \,\mathrm{softplus}(z_{u,\eta}) = m_\eta \log(1 + \exp z_{u,\eta}), \tag{46}$$

$$\lambda_u = \sum_{\eta \in \mathcal{D}_K} k_{u,\eta} B_{e,t}(i + \eta), \tag{47}$$

$$q_u(\Delta) = (1 - B_{e,t}(i))\{1 - \exp(-\Delta \lambda_u)\}, \qquad \Delta \geq 0 \text{ in hours.} \tag{48}$$

The neural formulas use numerical rate values expressed in h^{-1} and numerical elapsed durations expressed in hours. Equivalently, the physical rate corresponding to a softplus output is that output times 1 h^{-1}. Each $k_{u,\eta}$ has units of inverse hours per burned neighboring cell, so λ_u is an inverse-hour rate and $\Delta\lambda_u$ is dimensionless. All fitting examples have $B_{e,t}(i) = 0$ and $\Delta = 1$. The implementation evaluates $1 - e^{-x}$ as $-\operatorname{expm1}(-x)$ for numerical accuracy near zero. Its training output is $q_u(1)$, not λ_u and not a thresholded classification.

The exponential formula has a constant-rate survival interpretation. If $S_u(v)$ is the conditional probability that an initially unburned target has not transitioned by elapsed time v, and one assumes

$$\frac{dS_u(v)}{dv} = -\lambda_u S_u(v), \qquad S_u(0) = 1, \tag{49}$$

then separation of variables gives $S_u(v) = e^{-v\lambda_u}$ and transition probability $1 - S_u(\Delta) = q_u(\Delta)$. This is a modeling assumption used to motivate the link, not a fact established by the reconstruction. During one call, the current burned map and environmental kernel are held fixed; the code does not let newly burned cells become new sources within that call. Although the low-level formula accepts other nonnegative durations, the saved experiments fit and validate only the one-hour endpoint. Good behavior at a different duration is not demonstrated by these experiments.

At fixed environmental context, with the target held currently unburned, λ_u and q_u cannot decrease when a known off-center source changes from unburned to burned, because all contributions are nonnegative. If every source is unburned, both values are exactly zero. These constraints express the chosen architecture and do not prove a causal effect of any individual environmental channel.

8.3 Signed direct-convolution model

The separate direct experiment removes the coefficient nonnegativity constraint and the exponential output link:

$$w^{\mathrm{ker}}_{u,\eta} = m_\eta z_{u,\eta} \in \mathbb{R}, \tag{50}$$

$$s_u = (1 - B_{e,t}(i)) \sum_{\eta \in \mathcal{D}_K} w^{\mathrm{ker}}_{u,\eta} B_{e,t}(i + \eta), \tag{51}$$

$$p_u = \mathrm{clip}_{[0,1]}(s_u) = \min\{1, \max\{0, s_u\}\}. \tag{52}$$

For $u = (e, t, i)$, the local clipped output p_u is the global next-hour new-burn probability $q_i(t)$. It is distinct from the active-burning marginal $p_i(v)$ in the heat calculation. Here w^{ker} is a dimensionless kernel coefficient, distinct from the inverse inclusion weight w_u. The raw score s_u can be negative or exceed one and has no hazard-rate interpretation. The fitted signed model trains against s_u with an identity link. It applies no softplus, sigmoid, exponential conversion, or clipping in the training objective. The separate probability field p_u is used for checkpoint selection and all reported probability metrics. The direct model rejects durations other than exactly one hour. Adding a burned source can decrease s_u when its signed coefficient is negative, so monotonicity in burned-source count is not a property of this model.

8.4 Availability is an output of inference

Numerical zeros used internally to clean invalid inputs are accompanied by availability masks. They do not turn missing observations into known unburned cells. The original low-level sum can disregard an unknown source only when its coefficient is exactly zero; otherwise every contributing source must be known. Its generator requires all selected environmental channels throughout the $K \times K$ context. The direct model requires the complete burned-source context, including positions with a zero coefficient.

The boundary mode `unknown` invalidates required outside-grid context. Opting explicitly into `zero` assumes zero outside-grid features in the coordinates passed to the low-level model and no exterior burned-source contribution. Saved wrappers normalize first: exterior zero in normalized feature coordinates corresponds to each channel's raw training mean, which need not be zero in physical units. The training caches use complete interior contexts under `unknown`. A known already-burned target has zero new-burn probability even when its rate or environmental context is unavailable. For the original low-level model, a known target also has zero new-burn probability at zero duration even when its rate is unavailable. Probability validity and rate validity are therefore different objects.

The saved contact-only wrappers impose the fitted domain again at prediction: an unburned target without a known burned neighbor is returned as unavailable with a false validity mask and a NaN value, rather than accepted as a validated zero-probability forecast. The algebraic low-level zero in Equation (34) and this fitted-wrapper availability rule must not be confused.

9 Initialization, fitting objective, gradients, and model selection

9.1 Train-only baseline initialization

For the original family, the trainer first fits a constant nonnegative per-neighbor rate r on the native training sample:

$$\widehat{r}_{\mathrm{exp}} \approx \arg\min_{0\le r\le 20} \frac{1}{W_{\mathrm{tr}}} \sum_{u\in\mathcal{S}_{\mathrm{tr}}} w_u\{1-e^{-rn_{u,K}}-Y_u\}^2. \tag{53}$$

A bounded scalar numerical search fits this baseline; the code explicitly compares candidate endpoints. For larger supports it scales the search bracket by the smallest occupied positive neighbor count to avoid a saturated initial search, while retaining the original admissible endpoint candidates. This numerical procedure is not a proof that every possible nonconvex scalar objective attains its global minimum at the returned rate. The final-layer weights are set to zero and every output bias is set to the inverse softplus of $\max(\widehat{r}_{\mathrm{exp}}, 10^{-8})$. Consequently, before training every off-center coefficient equals this common initialized rate.

For the signed family, a constant linear coefficient has a closed-form training-only least-squares solution. Let $n_u = n_{u,K}$. Differentiating

$$F(r) = \sum_{u\in\mathcal{S}_{\mathrm{tr}}} w_u(rn_u - Y_u)^2 \tag{54}$$

gives $F'(r) = 2\sum_u w_u n_u(rn_u - Y_u)$. Since $\sum_u w_u n_u^2 > 0$ on a nonempty contact-only sample, the unique minimizing coefficient is

$$\widehat{r}_{\mathrm{lin}} = \frac{\sum_u w_u n_u Y_u}{\sum_u w_u n_u^2}. \tag{55}$$

The allowed coefficient is signed, although this particular baseline is nonnegative because its numerator is nonnegative. No clipping is used to fit it. The direct model initializes final-layer weights to zero and output biases to $\widehat{r}_{\mathrm{lin}}$, then masks the center coefficient to zero.

9.2 Weighted squared error, with two fixed scaling factors

Write $N_{\mathrm{tr}} = |\mathcal{S}_{\mathrm{tr}}|$, and define the global mean training weight and weighted training prevalence by

$$\overline{w}_{\mathrm{tr}} = \frac{W_{\mathrm{tr}}}{N_{\mathrm{tr}}}, \qquad \rho_{\mathrm{tr}} = \frac{\sum_{u\in\mathcal{S}_{\mathrm{tr}}} w_u Y_u}{W_{\mathrm{tr}}}. \tag{56}$$

The trainers require $0 < \rho_{\mathrm{tr}} < 1$. On a minibatch $\mathcal{B}$ of b examples, the exact mathematical objective before float32 rounding is

$$L_{\mathcal{B}}(\theta) = \frac{1}{b\rho_{\mathrm{tr}}} \sum_{u \in \mathcal{B}} \frac{w_u}{\overline{w}_{\mathrm{tr}}} \{f_\theta(u) - Y_u\}^2, \quad f_\theta(u) = \begin{cases} q_u(1), & \text{original family,} \\ s_u, & \text{signed direct family.} \end{cases} \tag{57}$$

Both scaling factors are fitted once from the full native training partition; neither is recalculated within a minibatch. In particular, the denominator does not sum minibatch weights. With uniformly shuffled training identities, the expectation of a fixed-size randomly selected batch loss, conditional on the prepared sample and fixed parameters, equals

$$\frac{1}{\rho_{\mathrm{tr}} W_{\mathrm{tr}}} \sum_{u \in \mathcal{S}_{\mathrm{tr}}} w_u \{f_\theta(u) - Y_u\}^2. \tag{58}$$

Thus the two positive global scaling factors change gradient magnitudes but not the minimizers of the unregularized full-sample weighted squared error. The sampling weights are not replaced by an arbitrary class-balancing loss. The optimizer separately uses a recorded weight-decay coefficient.

For an ideal population and unrestricted function class, squared loss has conditional mean $\mathbb{E}[Y \mid \text{model inputs, eligibility}]$ as its minimizer: if $m = \mathbb{E}[Y \mid X]$, then

$$\mathbb{E}[(f - Y)^2 \mid X] = (f - m)^2 + \mathrm{Var}(Y \mid X). \tag{59}$$

This explains the probability interpretation of fitting a binary response with squared error. The finite neural architecture, selected training sample, regularization, optimizer, and retrospective label definition still constrain what the fitted output estimates. Neither this identity nor a low Brier score establishes calibration of a fitted model.

9.3 Gradients show exactly which quantity is fitted

For $\alpha_u = 2w_u/(b\overline{w}_{\mathrm{tr}}\rho_{\mathrm{tr}})$, a batch example has derivative $\partial L/\partial f_u = \alpha_u(f_u - Y_u)$. In the original family, on a currently unburned valid target, the chain rule gives

$$\frac{\partial q_u(\Delta)}{\partial \lambda_u} = \Delta e^{-\Delta\lambda_u}, \tag{60}$$

$$\frac{\partial \lambda_u}{\partial z_{u,\eta}} = m_\eta B_{e,t}(i + \eta)\,\mathrm{sigmoid}(z_{u,\eta}), \tag{61}$$

$$\frac{\partial L}{\partial z_{u,\eta}} = \alpha_u(q_u - Y_u)\Delta e^{-\Delta\lambda_u} m_\eta B_{e,t}(i + \eta)\,\mathrm{sigmoid}(z_{u,\eta}). \tag{62}$$

The actual fitting duration is $\Delta = 1$. Hence the loss acts on the linked probability while gradients pass through its rate. It is not squared error between rate and target.

For the signed identity-link family,

$$\frac{\partial L}{\partial z_{u,\eta}} = \alpha_u(s_u - Y_u) m_\eta B_{e,t}(i + \eta). \tag{63}$$

A negative raw score on a positive target still gives a nonzero training gradient when a burned source contributes. If clipping had been placed inside the objective, its derivative outside $(0, 1)$ would instead be zero almost everywhere. The code intentionally trains the unclipped score. In both models, an inactive source or the fixed center yields zero derivative through that coefficient for this example. Backpropagation through the shared hidden layer uses

$$\mathrm{SiLU}'(a) = \mathrm{sigmoid}(a) + a\,\mathrm{sigmoid}(a)\{1 - \mathrm{sigmoid}(a)\}. \tag{64}$$

For example, if $g_{u,\eta} = \partial L/\partial z_{u,\eta}$, then the data-loss gradients are

$$\frac{\partial L}{\partial V_{\eta,h}} = \sum_{u\in\mathcal{B}} g_{u,\eta} H_{u,h}, \qquad \frac{\partial L}{\partial d_\eta} = \sum_{u\in\mathcal{B}} g_{u,\eta}, \tag{65}$$

$$\frac{\partial L}{\partial U_{h,c,\delta}} = \sum_{u\in\mathcal{B}} \left(\sum_{\eta} g_{u,\eta} V_{\eta,h} \right) \mathrm{SiLU}'(a_{u,h}) Z_{u,c,\delta}. \tag{66}$$

These formulas describe the differentiable complete-patch training path. They do not differentiate through availability masks or the discrete process that constructs reconstructed labels.

9.4 The Adam update used by the trainers

Let $\theta^{(k-1)} \in \mathbb{R}^{P_K}$ be the flattened parameter vector before minibatch update k, and let $g^{(k)} = \nabla_\theta L_{\mathcal{B}_k}$ be the data-loss gradient above. The current trainers use PyTorch 2.5.1 `torch.optim.Adam`, with learning rate $\eta = 10^{-3}$ and coupled weight decay $\lambda_{\mathrm{wd}} = 10^{-4}$. No optimizer-specific defaults are overridden for the moment coefficients or denominator constant, so $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon_{\mathrm{Adam}} = 10^{-8}$. These defaults were checked in the locally installed pinned optimizer source.

For each parameter coordinate, the mathematical update is

$$\widetilde{g}^{(k)} = g^{(k)} + \lambda_{\mathrm{wd}} \theta^{(k-1)}, \tag{67}$$

$$m^{(k)} = \beta_1 m^{(k-1)} + (1-\beta_1)\widetilde{g}^{(k)}, \tag{68}$$

$$v^{(k)} = \beta_2 v^{(k-1)} + (1-\beta_2)(\widetilde{g}^{(k)})^2, \tag{69}$$

$$\widehat{m}^{(k)} = m^{(k)}/(1-\beta_1^k), \qquad \widehat{v}^{(k)} = v^{(k)}/(1-\beta_2^k), \tag{70}$$

$$\theta^{(k)} = \theta^{(k-1)} - \eta \frac{\widehat{m}^{(k)}}{\sqrt{\widehat{v}^{(k)}} + \epsilon_{\mathrm{Adam}}}, \tag{71}$$

starting from $m^{(0)} = v^{(0)} = 0$. Products, powers, square roots, and division are componentwise. The index k counts minibatch updates, not epochs. Coupled weight decay adds the derivative of $\lambda_{\mathrm{wd}}\|\theta\|_2^2/2$ to the data gradient before the moment calculations; it is not an independent multiplicative shrinkage step. The first and second moment recurrences and their bias corrections adapt the step to the accumulated gradient history. Floating-point and backend execution can give small differences from these real-arithmetic equations.

The trainers clear old gradients for each batch, evaluate the appropriate linked probability or raw score, reject a nonfinite loss, backpropagate, reject any nonfinite parameter gradient, and perform the optimizer update. They do not clip gradients in the maintained fitting loop. Validation and test calculations do not update the optimizer or preprocessing statistics.

9.5 Epoch selection precedes test evaluation

The saved runs use Adam, learning rate 10^{-3}, weight decay 10^{-4}, batch size 2,048, hidden width 16, seed 7, at most 30 epochs, and patience 6, with float32 model calculations on the recorded MPS device. Each epoch shuffles training examples, performs the minibatch updates, and computes native weighted validation Brier. A checkpoint is saved only on a strict improvement, and training stops after six successive epochs without one. The original family selects by q_u; the direct family selects by $\mathrm{clip}(s_u, 0, 1)$. The direct raw squared error used for optimization and clipped squared error used for selection are distinct quantities. Clipping cannot increase squared error against a binary response, since projecting onto $[0,1]$ cannot increase distance to either 0 or 1.

The selected checkpoint is reloaded before the trainer first loads the test cache. Selected epochs for $K = 3, 5, 7, 9$ were respectively $2, 8, 5, 3$ in the original family and $4, 1, 4, 6$ in the

signed family. Optimization is finite and nonconvex; no guarantee of a global neural-network minimum is claimed.

The optimized center-patch computation is algebraically the center of the public full-spatial forward calculation. The trainers verify this agreement on up to 32 examples in float32 and independently with a CPU float64 model. Separate float32 tolerances account for different accumulation orders. Reloaded checkpoints also carry the ordered channel schema, normalization, one-hour target, zero-center constraint, split assignments, contact rule, and prepared-metadata hash. Prediction therefore has an explicit saved contract.

10 A common cohort for comparing kernel sizes and model families

10.1 Native populations are different

Increasing K changes both the complete-context requirement and the contact requirement. More cells lose full context near a missing-data boundary, while more distant burned sources can satisfy contact. Native eligible populations and their sampling weights consequently differ. Even the training-only normalization distribution differs. Comparing native metrics as though the models had been evaluated on the same examples is invalid.

The preserved comparison takes the native 9×9 prepared sample and retains only anchors having current burned contact inside its centered 3×3 support:

$$\mathcal{S}_{\text{common}} = \{u \in \mathcal{S}_9^+ : n_{u,3} > 0\}. \tag{72}$$

The complete 9×9 patch guarantees that each smaller centered patch is complete. Membership depends on current context and current burned state; it does not use the next-hour target. Every model receives the same event, cutoff, row, column, label, and original 9×9 inverse inclusion weight. For $K < 9$, the model reads only the centered $K \times K$ crop and applies its own saved normalization. There is no intersection of independently sampled negative rows and no refitting on the common evaluation cohort.

Common partition	Fires	Samples	Positives	$\sum w$	Prevalence
Training	22	100,259	64,285	2,291,049.6466	0.0280592
Validation	3	12,143	6,579	343,561.4514	0.0191494
Test	8	16,807	7,378	85,133.3236	0.0866641

The represented populations are weighted estimates of the complete 9×9, current 3×3-contact subpopulation. The common validation cohort contributes only three fires, all in the United States; no Korean fire survives its complete 9×9 context and centered 3×3 contact requirements. The common training evaluation covers two Korean and twenty US fires, and common test covers two Korean and six US fires. Kernel-size selection therefore has direct common-validation evidence only from the retained US fires, despite fitting pooled-country models. No uncertainty intervals or independent fire-level resampling were supplied with these results.

10.2 Kernel selection is a separate validation operation

Each already fitted kernel keeps the epoch selected on its own native validation population. The common validation population then selects the kernel size by minimum weighted Brier, breaking exact ties toward the smaller kernel. The choice is saved before the common test cache is read. Both families selected 5×5. The shared validation Brier values are:

Family	3×3	5×5	7×7	9×9
Original	0.01868646	0.01859248	0.01869454	0.01867230
Signed direct, clipped	0.01869695	0.01859025	0.01868570	0.01865468

Kernel geometry, training population, parameter count, and fitted normalization all vary across supports. This comparison tests the resulting trained systems on a common evaluation population; it is not an isolated, equal-capacity ablation of spatial radius.

11 Metrics, saved numerical results, and failure modes

11.1 Weighted Brier and exact Bernoulli log loss

Sampling weights w_u are indexed by examples u and are distinct from physical dry fuel loading w_i indexed by cells i. For evaluation examples $u \in \mathcal{S}$ with positive weights, labels $Y_u \in \{0,1\}$ and probability outputs $p_u \in [0,1]$, put $W = \sum_u w_u$. The weighted Brier score is

$$\widehat{\mathrm{BS}} = \frac{1}{W}\sum_u w_u (p_u - Y_u)^2. \tag{73}$$

The original model supplies $p_u = q_u(1)$; the direct model supplies the clipped view from Equation (52). The direct raw-score mean squared error is recorded separately and must not be called a probability Brier score without explaining that scores may lie outside $[0,1]$.

Define the Bernoulli negative log-likelihood contribution without an ambiguous $0 \log 0$ expression:

$$\ell(Y,p) = \begin{cases} -\log p, & Y = 1, \\ -\log(1-p), & Y = 0, \end{cases} \qquad -\log 0 = +\infty. \tag{74}$$

The exact weighted mean is $\sum_u w_u \ell(Y_u, p_u)/W$. A single positive target with $p_u = 0$, or negative target with $p_u = 1$, makes it infinite. This does not imply infinite squared error. It is a support failure under the fitted Bernoulli probabilities.

The code additionally reports a diagnostic finite log loss using

$$p_u^{(\epsilon)} = \min\{1-\epsilon, \max\{\epsilon, p_u\}\}, \qquad \epsilon = 10^{-7}. \tag{75}$$

Its weighted log loss uses $p_u^{(\epsilon)}$ in place of p_u. The finite diagnostic is not the exact NLL and does not change the fitted model or its unmodified output probabilities.

11.2 Weighted ranking metrics, with explicit tie treatment

Let $P = \sum_u w_u Y_u$ and $N = \sum_u w_u (1 - Y_u)$. Sort distinct scores in descending order, and let τ_j be the jth distinct threshold. All examples tied at a threshold enter simultaneously. Define

$$\mathrm{TP}_j = \sum_{u: p_u \geq \tau_j} w_u Y_u, \qquad \mathrm{FP}_j = \sum_{u: p_u \geq \tau_j} w_u (1 - Y_u), \tag{76}$$

$$R_j = \mathrm{TP}_j / P, \qquad F_j = \mathrm{FP}_j / N, \tag{77}$$

$$\Pi_j = \mathrm{TP}_j / (\mathrm{TP}_j + \mathrm{FP}_j). \tag{78}$$

With $R_0 = F_0 = 0$, the implemented average precision and trapezoidal ROC area are

$$\widehat{\mathrm{AP}} = \sum_j (R_j - R_{j-1}) \Pi_j, \qquad \widehat{\mathrm{AUC}} = \sum_j (F_j - F_{j-1}) \frac{R_j + R_{j-1}}{2}. \tag{79}$$

AP is undefined when there are no weighted positives; ROC AUC additionally requires weighted negatives. The code returns unavailable values in these cases. Ranking measures do not test probability calibration, and clipping can introduce ties that affect ranking.

11.3 Fixed-bin reliability

The descriptive reliability calculation fixes bin edges in advance at

$$0,\ 0.01,\ 0.02,\ 0.05,\ 0.10,\ 0.20,\ 0.50,\ 1. \tag{80}$$

Each bin is left closed and right open, except the final bin includes one. For a nonempty bin b, define

$$W_b = \sum_{u \in b} w_u, \qquad \overline{p}_b = \frac{\sum_{u \in b} w_u p_u}{W_b}, \qquad \overline{y}_b = \frac{\sum_{u \in b} w_u Y_u}{W_b}. \tag{81}$$

The recorded weighted expected calibration error is

$$\widehat{\mathrm{ECE}} = \sum_{b: W_b > 0} \frac{W_b}{W} |\overline{p}_b - \overline{y}_b|. \tag{82}$$

It is a fixed-bin descriptive statistic, sensitive to the stated bins and population. It provides no confidence interval and no general calibration guarantee. The saved metric calculations are in `wildfire_burned_area_hourly/probability_model/metrics.py`.

11.4 Saved common test results

The following table reproduces the saved common test comparison, rounded for readability. Every row uses the same 16,807 identities, 7,378 positive labels, and weights. "Zero positive" counts realized positive labels given exact probability zero.

Family	K	Brier	AP	ROC AUC	Floor LL	Exact NLL	Zero-pos.
Original	3	0.067219	0.383729	0.831045	0.242273	0.242273	0
Original	5	0.069587	0.333213	0.825043	0.249690	0.249690	0
Original	7	0.071997	0.387380	0.838929	0.259517	0.259517	0
Original	9	0.068417	0.363097	0.829479	0.251728	0.251728	0
Signed direct	3	0.069701	0.432513	0.823782	0.274618	$+\infty$	164
Signed direct	5	0.064982	0.471449	0.842467	0.262619	$+\infty$	215
Signed direct	7	0.067309	0.429701	0.840482	0.244700	$+\infty$	61
Signed direct	9	0.077888	0.289826	0.767516	0.329989	$+\infty$	290

No common test row had a negative label assigned probability one. Thus the infinite exact NLL of every signed direct model is caused by positive labels receiving probability zero. In these models, negative raw scores are clipped to zero; contact conditioning does not prevent this different support failure.

For the selected direct 5×5 model, saved common test raw scores range from -0.0415605344 to 0.7262458205. There are 2,162 negative raw scores, or 12.8637% of sampled targets and 19.5557% of represented population weight; no raw score exceeds one. Of the 7,378 positive targets, 215 receive zero probability. Its raw weighted squared error is 0.0650427772 and its clipped Brier is 0.0649824060. Mean clipped probability is 0.0458195, below weighted outcome prevalence 0.0866641. Its fixed-bin ECE is 0.0408446, compared with 0.0379574 for the original 5×5 model. These results do not support describing the direct model as calibrated.

The selected direct 5×5 improves common test Brier and ranking relative to the original 5×5, but worsens its floor log loss and has infinite exact NLL. The direct 3×3 and 9×9 have worse Brier than their same-size original counterparts, while 5×5 and 7×7 improve it. Therefore the experiment does not establish uniform superiority of the signed family. Also, although both validation selections chose 5×5, the original 3×3 has lower descriptive test Brier than the selected original 5×5; the kernel choice is not revised using that test observation.

11.5 Result files and reproducibility boundaries

The primary saved evidence is `probability_model/training_runs/20261007_1h_kernel_sweep/common_metrics.json` and `probability_model/training_runs/20261007_1h_direct_convolution_signed/common_metrics.json` relative to `wildfire_burned_area_hourly/`. Associated `common_predictions` archives retain labels, weights, cell-hour identities, per-kernel neighbor counts, and probabilities; direct archives also retain unclipped raw scores. The saved protocols declare kernel sizes, source/cache identities, training parameters, clipping behavior, and validation-only selection. The direct evaluation code checks exact equality of all shared identities, labels, weights, and neighbor counts against the preceding original common-cohort archives.

The saved common test public-wrapper versus center-computation checks have maximum absolute discrepancies no greater than 1.79×10^{-7} for the original family and 5.97×10^{-8} for the signed family, below their respective recorded float32 tolerances. These are computational agreement checks. They do not validate the reconstructed label against an independent observation or quantify generalization uncertainty. The experiments use one fixed sampling seed and one fitted run per configuration; the reported comparisons are retrospective, population-conditioned, and descriptive.

12 Step 5: ecological fuel loading and its coverage contract

12.1 Two different reference catalogs have two different roles

The radiation extension uses FCCS dry loading to prescribe a conditional heat source, whereas the residence extension uses the Scott–Burgan FBFM40 catalog to prescribe a passing-front duration. These are separate inputs. A code in one catalog is not a code in the other catalog. A biome class is neither a dry mass nor a particle diameter. No conversion from a biome label, canopy height, relative humidity, or soil moisture to dry fuel mass is implemented by `EcologicalFuelMaps`. The implemented contracts are in `expected_heat_flux/ecology.py` and `expected_heat_flux/residence.py`.[6]

LANDFIRE assigns vegetation to reference FCCS fuelbeds. The reference loading table describes fuelbed attributes, rather than a field measurement of the fuel available on the date of a particular fire.[7] FCCS represents canopy, shrub, herbaceous, woody, litter–lichen–moss, and ground-fuel strata; a reference fuelbed may need customization for an actual site.[8] The package reads the previously prepared total-loading raster. It does not implement all FCCS fire-behavior, consumption, or emissions calculations.

12.2 Objects, units, and conversion to SI

Let i denote one cell of a target raster with shape $[H, W]$, where H and W are positive integer numbers of rows and columns. Let w_i^{mapped} be the mean reference dry loading over the mapped portion of cell i, in $\mathrm{kg\,m^{-2}}$ of that portion, and let $c_i \in [0, 1]$ be the fraction of the cell described by the mapped loading. An unreadable loading or coverage is represented as missing, not as zero. The distinction between a mapped-part mean and a whole-cell mean is essential: if $c_i = 1/2$, then w_i^{mapped} is the mean over half of the cell. Multiplication by c_i yields its contribution per unit *whole-cell* area.

The raster loading is interpreted as U.S. short tons per international acre. One U.S. short ton is 2000 international pounds, one international pound is exactly 0.45359237 kg, one international acre is 43560 square international feet, and one international foot is exactly 0.3048 m. Hence the

implemented conversion factor is

$$C_w = \frac{2000 \times 0.45359237}{43560 \times (0.3048)^2} \frac{\mathrm{kg\,m^{-2}}}{\mathrm{short\ ton\,acre^{-1}}}$$
$$= 0.2241702312388912 \frac{\mathrm{kg\,m^{-2}}}{\mathrm{short\ ton\,acre^{-1}}}, \qquad w_i^{\mathrm{mapped}} = C_w L_i, \tag{83}$$

where L_i is the loading quantity read from the raster, expressed in U.S. short tons per international acre; the stored scalar is its numerical magnitude in those units. This is a unit conversion; it supplies no information about the event's actual consumed mass.

12.3 Alignment, physical validity, and complete coverage

The loading and coverage rasters must each have one static band. Their shapes, coordinate reference systems (CRSs), and six affine-transform coefficients must agree exactly. Their CRS must be projected with a linear unit equal to one metre. If a prediction grid or an expected shape is supplied, the reader checks that alignment as well. Shape equality by itself is insufficient: a shifted grid would associate the loading with a different location.

The reader replaces a negative or nonfinite loading by missing data. It also replaces a nonfinite coverage or a coverage outside $[0, 1]$ by missing data. The numerical tolerance for declaring coverage complete is a caller-selected $\varepsilon_c \in [0, 1)$, whose default is 10^{-6}. The reader's complete-input indicator is

$$V_i^{\mathrm{fuel}} = \mathbf{1}\{w_i^{\mathrm{mapped}} \text{ is finite}\}\mathbf{1}\{c_i \text{ is finite}\}\mathbf{1}\{c_i \geq 1 - \varepsilon_c\}. \tag{84}$$

Here $\mathbf{1}\{\mathcal{A}\}$ equals one if statement $\mathcal{A}$ is true and zero otherwise. A finite, nonnegative loading of zero is legitimate when it was actually supplied with valid coverage; missing loading is a different state. The accepted tolerance concerns numerical coverage. It does not quantify uncertainty in fuel mass. Setting $\varepsilon_c = 0$ requires exact coverage $c_i = 1$.

Without an explicit fallback loading, the conversion to combustion parameters requires every cell to satisfy (84); otherwise it fails. If a coverage just below one is accepted within tolerance and no fallback is given, the implementation uses w_i^{mapped} directly as the whole-cell loading. This is the precise numerical tolerance convention used by the code.

12.4 Explicit completion of the unmapped fraction

Let $w_i^{\mathrm{fallback}} \geq 0$ be a finite loading supplied by the caller for the *unmapped* portion, in $\mathrm{kg\,m^{-2}}$ of that portion. A scalar fallback applies to every cell; an array must broadcast to the existing spatial shape $[H, W]$, without introducing a batch or time dimension. When a fallback is supplied, the whole-cell dry loading is

$$w_i = \begin{cases} c_i w_i^{\mathrm{mapped}} + (1 - c_i) w_i^{\mathrm{fallback}}, & w_i^{\mathrm{mapped}} \text{ and } c_i \text{ are readable,} \\ w_i^{\mathrm{fallback}}, & \text{otherwise.} \end{cases} \tag{85}$$

For a cell of area $|S_i|$ in $\mathrm{m^2}$, the first case follows from adding the dry masses $|S_i| c_i w_i^{\mathrm{mapped}}$ and $|S_i|(1 - c_i) w_i^{\mathrm{fallback}}$, then dividing by $|S_i|$. If $c_i = 1$, the fallback contributes nothing. If $c_i = 0$, the fallback supplies the whole-cell loading. If either mapped input is unreadable, the implementation uses the fallback for the entire cell; it cannot recover a known mapped contribution from unreadable data.

For example, if $c_i = 0.4$, $w_i^{\mathrm{mapped}} = 2$, and $w_i^{\mathrm{fallback}} = 10$ in $\mathrm{kg\,m^{-2}}$, then $w_i = 0.4(2) + 0.6(10) = 6.8\,\mathrm{kg\,m^{-2}}$. Replacing the whole cell by the fallback would discard the mapped information; using the mapped mean alone would leave the unmapped fraction unspecified.

Under the *reference-input assumptions*, and assuming only that the unmapped loading is nonnegative, a mathematical lower bound is

$$w_i^{\text{lower}} = \begin{cases} c_i w_i^{\text{mapped}}, & \text{both mapped inputs are readable,} \\ 0, & \text{otherwise.} \end{cases} \tag{86}$$

Indeed, the unknown nonnegative mass contributes a nonnegative summand in (85). This is not a lower bound on true field fuel mass, because the reference loading itself can be inaccurate. The zero in the second case is the trivial nonnegative bound, not an observed zero loading.

12.5 Dry fuel, moisture, and prescribed combustion parameters

For a fuel sample with dry mass $m_{\text{dry}} > 0$ and water mass $m_{\text{water}} \geq 0$, dry-basis moisture would be $M = m_{\text{water}}/m_{\text{dry}}$ (dimensionless), or $100M$ percent. That physical definition is distinct from relative humidity, soil-water content, and a categorical biome label. The ecological reader neither derives dead-fuel moisture nor estimates live-fuel moisture. Weather and fuel-moisture fields can be predictor covariates, but the heat extension does not automatically turn them into a moisture correction of dry loading, combustion heat, consumed fraction, or residence time.

After specifying w_i, the caller must also specify a positive effective heat of combustion $H_{c,i}$ in $\mathrm{J\,kg^{-1}}$, a positive combustion-phase duration T_i in seconds, and a consumed dry-mass fraction $f_i \in [0, 1]$. The conditional uniform heat-release-rate density is

$$Q_i = \frac{w_i f_i H_{c,i}}{T_i} \quad [\mathrm{W\,m^{-2}}]. \tag{87}$$

The effective heat is supplied, rather than inferred as a moisture-corrected gross calorific value. If T_i is a main-front flaming duration, f_i must refer to dry fuel consumed in that modeled phase; it must not silently mean all fuel eventually consumed by residual flaming and smouldering. A separate radiative fraction is required by the radiation operator.

The completed integrated FishCreek demonstration prescribes $H_c = 18 \times 10^6\,\mathrm{J\,kg^{-1}}$, $T = 300\,\mathrm{s}$, $f = 0.10$, an unmapped-loading fallback of $10\,\mathrm{kg\,m^{-2}}$, and radiative fraction 0.30. These are illustrative scenario parameters in its saved summary. They are not estimated from observations of FishCreek combustion. The 300 s duration in that demonstration is separate from the FBFM40 passing-front durations used by the active-flame demonstration described below.

13 Fuel-dependent surface-flame residence scenarios

13.1 What duration is being assigned

In this section, duration is denoted τ_i^{fl} to distinguish it from a separately supplied heat-emission duration T_i. At a representative grid location i, let $\tau_i^{\text{fl}} > 0$ be a deterministic, finite duration in seconds. Anderson's passing-front approximation and its Andrews formulation describe the time for a surface-fire flaming front to pass one fixed fuel location. The formulation excludes the indefinite trailing edge and assumes no effect of fuel-bed packing.[9, 10] It is distinct from the time since ignition, the time to traverse a coarse cell, the interval during which some part of that cell contains any flame, residual-flame extinction, crown-fire duration, or smouldering duration.

The object `FlameResidenceTime` stores either one scalar duration or a complete matrix of shape $[H, W]$, plus immutable provenance. A scalar declares the same scenario duration at all locations to which it is applied. A matrix must align with the consumer's spatial grid. Zero, negative, nonfinite, masked, boolean, string, and complex durations are rejected; a missing duration is not assigned zero. The class copies the input and freezes its duration and provenance. Direct construction requires a provenance mapping.

Assigning a point-scale τ_i^{fl} to a 375 m grid cell is an additional spatial modeling assumption. It assigns a duration to a representative location; it does not resolve the arrival progression or flame geometry within the footprint. A catalog assignment supplies a reference-fuel scenario, not an event-specific measurement of duration.

The heat-emission duration T_i in (87) and the flame residence τ_i^{fl} are separate objects. The helper `CombustionParameters.from_flame_residence` explicitly sets $T_i = \tau_i^{\mathrm{fl}}$ when constructing combustion parameters from the residence scenario. When `WildfireHeatModel` is also supplied a `residence_time`, it enforces this equality and rejects a mismatch. The equality is a chosen bridge between modeled phases, not a conclusion that all reference fuel is consumed during front passage; the consumed fraction must describe that same phase. Without this explicit bridge, an independently supplied heat duration and a flame residence must retain their distinct meanings.

13.2 From particle diameter: the Anderson relation

Let $d_i > 0$ be the particle diameter in metres. Define the dimensionless original-unit numerical values $d_{\mathrm{inch}} = d_i/(0.0254\,\mathrm{m})$ and $\tau_{\mathrm{min}}^{\mathrm{fl}} = \tau_i^{\mathrm{fl}}/(60\,\mathrm{s})$. The implemented source law is $\tau_{\mathrm{min}}^{\mathrm{fl}} = 8d_{\mathrm{inch}}$. Converting units gives

$$d_{\mathrm{inch}} = \frac{d_i}{0.0254\,\mathrm{m}}, \tag{88}$$

$$\tau_i^{\mathrm{fl}} = 8\frac{\mathrm{min}}{\mathrm{inch}}\frac{d_i}{0.0254\,\mathrm{m/inch}}\frac{60\,\mathrm{s}}{\mathrm{min}} = C_d d_i, \quad C_d = 18\,897.63779527559\,\mathrm{s\,m^{-1}}. \tag{89}$$

Thus a particle of diameter one inch, $d_i = 0.0254\,\mathrm{m}$, gives $\tau_i^{\mathrm{fl}} = 480\,\mathrm{s}$. A particle of diameter $0.002\,\mathrm{m}$ gives $\tau_i^{\mathrm{fl}} = 37.79527559055118\,\mathrm{s}$. The law approximates a uniform fuel array; it does not establish that every fuel particle in a heterogeneous cell burns out by this time. The public Anderson constructor validates positivity and finiteness but does not impose a calibrated species-specific diameter range. Allowing a numerical input therefore does not validate its physical use.

13.3 From characteristic surface-area-to-volume ratio

Let $\sigma_i > 0$ be characteristic fuel-particle surface area divided by particle volume, in $\mathrm{m^{-1}}$. Its numerical value in $\mathrm{ft^{-1}}$ is 0.3048 times its numerical value in $\mathrm{m^{-1}}$. Andrews's residence relation is $\tau_{\mathrm{min}}^{\mathrm{fl}} = 384/\sigma_{\mathrm{ft^{-1}}}$.[10] The implemented SI form is consequently

$$\tau_i^{\mathrm{fl}} = \frac{C_\sigma}{\sigma_i}, \qquad C_\sigma = \frac{384 \times 60}{0.3048}\,\mathrm{s\,m^{-1}} = 75\,590.55118110236\,\mathrm{s\,m^{-1}}. \tag{90}$$

The unit check is $(\mathrm{s\,m^{-1}})/(\mathrm{m^{-1}}) = \mathrm{s}$. Using $384/\sigma_i$ with an SI input would omit both conversions.

For a cylinder of diameter d_i and length $\ell > 0$, neglecting its end area, the side area is $\pi d_i \ell$ and the volume is $\pi d_i^2 \ell/4$. Hence $\sigma_i = 4/d_i$, so

$$\frac{C_\sigma}{\sigma_i} = \frac{C_\sigma d_i}{4} = C_d d_i, \qquad \frac{384 \times 60}{4 \times 0.3048} = \frac{8 \times 60}{0.0254}. \tag{91}$$

The equality is exact because $0.3048 = 12(0.0254)$. This geometric argument depends on the long-cylinder approximation. It is not a general conversion from leaf geometry, tree height, or raster width to a particle diameter.

13.4 From the FBFM40 reference catalog

The catalog constructor supports exactly the 40 burnable standard numeric identifiers shown in Table 3. The components of each model are dead 1-hour, dead 10-hour, dead 100-hour, live

herbaceous, and live woody fuel. These time-lag class names classify fuel components; they are not assigned flaming durations. The code transcribes the dry loads and particle SAV values from Scott–Burgan table 7.[11]

Reference group	Supported numeric identifiers	Number
Grass (GR)	101–109	9
Grass–shrub (GS)	121–124	4
Shrub (SH)	141–149	9
Timber understory (TU)	161–165	5
Timber litter (TL)	181–189	9
Slash/blowdown (SB)	201–204	4

Table 3: The complete set of burnable FBFM40 codes accepted by the residence catalog constructor. Numeric gaps between groups are not additional supported models.

For one reference fuelbed, index its components by j. Let $w_j \geq 0$ be the dry loading in $\mathrm{kg\,m^{-2}}$, $\rho_j > 0$ the particle material density in $\mathrm{kg\,m^{-3}}$, and $\sigma_j > 0$ its particle SAV in $\mathrm{m^{-1}}$. The particle volume per unit ground area is w_j/ρ_j. Its surface area per unit ground area is consequently

$$a_j^{\mathrm{surf}} = \frac{w_j \sigma_j}{\rho_j}. \tag{92}$$

The ratio a_j^{surf} has units of fuel surface area per ground area, $\mathrm{m^2/m^2}$, hence is dimensionless; it is distinct from the footprint set S_j. Surface-area weighting means taking the arithmetic mean of the particle SAVs with weights proportional to these surfaces, not to dry loads alone:

$$\overline{\sigma} = \frac{\sum_j a_j^{\mathrm{surf}} \sigma_j}{\sum_j a_j^{\mathrm{surf}}} = \frac{\sum_j w_j \sigma_j^2/\rho_j}{\sum_j w_j \sigma_j/\rho_j}. \tag{93}$$

This collapses the live/dead category weighting and within-category surface weighting described by Andrews into one expression.[10] Scott–Burgan uses one common particle density of $32\,\mathrm{lb\,ft^{-3}}$, dead 10-hour SAV of $109\,\mathrm{ft^{-1}}$, and dead 100-hour SAV of $30\,\mathrm{ft^{-1}}$.[11] The common density and any common loading-unit conversion cancel, giving the implemented formula

$$\overline{\sigma}_{\mathrm{ft^{-1}}} = \frac{\sum_{j=1}^{5} w_j \sigma_{j,\mathrm{ft^{-1}}}^2}{\sum_{j=1}^{5} w_j \sigma_{j,\mathrm{ft^{-1}}}}, \qquad \overline{\sigma}_i = \frac{\overline{\sigma}_{\mathrm{ft^{-1}}}}{0.3048}\,\mathrm{m^{-1}}, \qquad \tau_i^{\mathrm{fl}} = \frac{384 \times 60}{\overline{\sigma}_{\mathrm{ft^{-1}}}}\,\mathrm{s}. \tag{94}$$

The subscript on a SAV numerical value declares its unit. Entries with zero loading contribute zero to both sums. In particular, the source-table SAV placeholder 9999 in a zero-load component is not physically weighted as fine fuel.

For GR1, only the dead 1-hour and live-herbaceous entries have positive load: $w = (0.10, 0.30)$ in short tons per acre, with particle SAVs $(2200, 2000)\,\mathrm{ft^{-1}}$. Thus

$$\overline{\sigma}_{\mathrm{ft^{-1}}} = \frac{0.10(2200)^2 + 0.30(2000)^2}{0.10(2200) + 0.30(2000)} = \frac{1\,684\,000}{820} = 2053.658536585366, \tag{95}$$

$$\tau^{\mathrm{fl}} = \frac{23040}{2053.658536585366}\,\mathrm{s} = 11.219002375296912\,\mathrm{s}. \tag{96}$$

The published rounded characteristic SAV, $2054\,\mathrm{ft^{-1}}$, is useful for checking the transcription, but the constructor calculates the weighted ratio from the component parameters before converting to duration.

The GS4 source has a recorded inconsistency: its component values in table 7 imply approximately $1631\,\mathrm{ft^{-1}}$, consistent with Andrews's table 8c, while Scott–Burgan's individual GS4 summary prints $1674\,\mathrm{ft^{-1}}$. The implementation consistently uses the component table calculation and records the discrepancy in provenance. It does not silently substitute the inconsistent summary value.

13.5 Why curing does not change this particular characteristic SAV

Let a live-herbaceous component have dry loading w_h, SAV σ_h, and density ρ_h. Let $\alpha \in [0, 1]$ be the fraction transferred to a new dead herbaceous component. Scott–Burgan preserves particle SAV during that transfer; it does not simply combine the transferred material with a dead 1-hour component of another SAV.[11] If the transfer also retains the common catalog density, then the contributions to the denominator of (93) are

$$\frac{(1-\alpha)w_h\sigma_h}{\rho_h} + \frac{\alpha w_h\sigma_h}{\rho_h} = \frac{w_h\sigma_h}{\rho_h}, \tag{97}$$

and the corresponding numerator contributions are

$$\frac{(1-\alpha)w_h\sigma_h^2}{\rho_h} + \frac{\alpha w_h\sigma_h^2}{\rho_h} = \frac{w_h\sigma_h^2}{\rho_h}. \tag{98}$$

Therefore both sums, $\overline{\sigma}$, and this SAV-based τ^{fl} are invariant to α. The constructor can compute this duration without choosing a live moisture or curing fraction. This algebraic invariance is specific to this duration calculation. It does not imply that moisture and curing have no effect on ignition, spread, heat release, or consumption. The package does not estimate a curing fraction from moisture as part of the residence constructor.

13.6 Unsupported, missing, and nonburnable codes

The input codes must be numeric integer-valued identifiers in a scalar or a nonempty spatial matrix. A finite noninteger code is rejected even when a fallback is supplied. Missing, masked, custom, and unsupported integer codes, including standard nonburnable codes $91, 92, 93, 98, 99$, have no residence assigned by the burnable catalog. If any such location occurs, the constructor fails unless the caller explicitly supplies a strictly positive, finite `fallback_sav_m_inv`. That fallback must be a scalar or match the code matrix shape exactly.

On a supported cell the catalog SAV is retained. On an unsupported or missing cell the fallback SAV is substituted and (90) is applied. Overriding a nonburnable class in this way is a caller's combustion scenario; it is not a duration attributed to the mapped class and is not evidence that the cell has combustible fuel. Provenance separately records catalog-cell, fallback-cell, and missing-code counts, finite unsupported codes, and the fallback value. No fallback is implicit in the public catalog constructor.

The active-flame example *does* explicitly set $\sigma_{\mathrm{fallback}} = 2000\,\mathrm{ft}^{-1} = 6561.679790026246\,\mathrm{m}^{-1}$, giving $\tau^{\mathrm{fl}} = 11.52\,\mathrm{s}$ on fallback locations. This is an example argument, not a universal residence-time default.

13.7 A distinct laboratory relation for jarrah round wood

`from_eucalypt_wood_diameter` implements Burrows's thesis equation 6-1:

$$\tau_i^{\mathrm{fl}} = 0.871 \left(\frac{d_i}{1\,\mathrm{mm}}\right)^{1.875}\ \mathrm{s}, \qquad 1\,\mathrm{mm} \le d_i \le 16\,\mathrm{mm}. \tag{99}$$

The dimensionless ratio in the power makes the unit convention explicit. This numerical law uses a *millimetre* input, while (89) uses a metre input. The constructor rejects extrapolation outside the closed experimental interval.

The original methods describe individual oven-dried jarrah (*Eucalyptus marginata*) round-wood particles cut to 10 cm, ignited over a pilot flame, then allowed to burn after the pilot was extinguished. The measured endpoint is flame-out, distinguished from cessation of glowing combustion.[12] This fuel-specific particle flame-out relation is separate from Anderson's passing-front approximation. Neither law is selected merely from a biome label. The thesis includes

still-air laboratory context and ambient temperatures of 26–32 °C in the surrounding experimental description; the code records these conditions as scenario metadata. Transferring the law to another species, wet fuel, wind, a whole fuelbed, or a coarse cell requires additional evidence that the present package does not supply.

The constants 0.871 and 1.875 are treated as fixed in the implemented scenario. Although the thesis prints regression standard errors, the package does not sample those parameters, propagate their uncertainty, or construct a flame-duration confidence interval. The exact implemented equation is supported by the 1994 thesis; the source register distinguishes this from the later 2001 article, whose full equation was not recorded as inspected.

13.8 Deterministic durations and the limits of uncertainty claims

There is no implemented residence-time probability distribution, moisture-based duration fit, quantile sampler, or Monte Carlo propagation of ecological parameter uncertainty in these modules. A duration fixed by a scenario can formally be represented by a point mass at τ_i^{fl}, but that mathematical representation is not a fitted uncertainty model. The within-hour uniform arrival assumption used by the forecast bridge concerns arrival time, not uncertainty in τ_i^{fl}. A spatial collection of different prescribed durations is also not a probability distribution of the unknown duration at a given cell.

Likewise, an arrival-time standard deviation alone does not identify an arrival distribution or its quantiles. A mean and variance can be shared by many distributions with different tail probabilities. None of the residence constructors converts a kriging standard-deviation raster into a normal, lognormal, or calibrated arrival law. Conditional source-strength assumptions and fuel uncertainty therefore remain outside the computed heat bounds unless an additional model is supplied explicitly.

13.9 Saved FishCreek scenario and verification scope

The active-flame demonstration uses the 2016 LANDFIRE modal FBFM40 class in each 375 m cell for a 2024 event. The saved grid has $H = 103$, $W = 73$, and $HW = 7519$ cells. It assigns 7346 durations from burnable catalog classes and 173 durations from the explicit fallback for codes 98 and 99. Its assigned durations have minimum 11.219002375296913 s, median 18.83071682602384 s, and maximum 18.83071682602384 s. Those numbers describe the saved reference-fuel scenario, not measured flame durations.

For the same event, 2642 saved valid arrival-standard-deviation values have minimum 32.776554107666016 h, median 87.61341857910156 h, and maximum 246.66561889648438 h. These exact raster values show that the reported arrival uncertainty is much larger than the selected point durations. The nominal current-flame map at the saved cutoff has an empty active-position selection. This outcome cannot establish absence of observed flames: the calculation uses reconstructed point arrivals and does not integrate their uncertainty or represent residual flames.

Current verification of the residence and ecological-reader contracts passed 118 selected tests. They check the original and SI unit equations, cylinder geometry, all 40 rounded characteristic SAVs, zero-load placeholders, immutable residence provenance, unsupported-code fallbacks, Burrows's tested range, metric raster alignment, incomplete coverage, nonnegative lower bounds, and the exact coverage-weighted fallback formula. These are mathematical and software checks. They are not observational calibration of duration, fuel consumption, active-flame positions, or wildfire heat exposure.

14 Step 6: activity conversion and expected incident radiant heat

This section documents the methods implemented in `expected_heat_flux/model.py`, `expected_heat_flux/temporal.py`, `expected_heat_flux/pipeline.py`, and `expected_heat_flux/flames.py`. The result is the expected *incident radiant irradiance* at specified receiver points, in $\mathrm{W\,m^{-2}}$, under an explicit combustion, timing, and geometry scenario. It is not an estimate of every component of wildfire heat transfer. In particular, convection, receiver absorption, temperature, conduction, terrain obstruction, smoke-dependent attenuation, and emission from outside the supplied source raster are absent from this implementation. The mathematical calculations can be verified without establishing that the physical assumptions accurately describe a particular wildfire.[4]

14.1 Objects, units, and the source-state interpretation

Let $\mathcal{J} = \{0, \dots, H-1\} \times \{0, \dots, W-1\}$ be the finite source-cell index set, where H, W are positive integers. Index $j = (r, c)$ denotes row r and column c, with both indices starting at zero. A projected affine grid in metres has coefficients (a, b, c_0, d, e, f_0) and pixel-corner map

$$(x, y) = (ac + br + c_0,\ dc + er + f_0). \tag{100}$$

The column and row edge vectors are $\mathbf{u} = (a, d)$ and $\mathbf{v} = (b, e)$, respectively. Each cell has horizontal ground area

$$A_{\text{cell}} = |ae - bd| > 0 \qquad [\mathrm{m^2}], \tag{101}$$

and centre

$$\mathbf{x}_j = (a(c + \tfrac{1}{2}) + b(r + \tfrac{1}{2}) + c_0, d(c + \tfrac{1}{2}) + e(r + \tfrac{1}{2}) + f_0). \tag{102}$$

This representation supports rotation, shear, rectangular spacing, and either row direction. Longitude and latitude are angular coordinates and cannot be used as metre coordinates. The radiation calculation requires a projected metric grid; `AffineGrid` itself does not inspect a coordinate-system identifier. The caller or data adapter must establish that requirement.

Let $\mathcal{F}_0$ denote the information used at forecast time zero. For cell j, let $Z_j(t) \in \{0, 1\}$ indicate active burning at relative time t seconds, and let

$$p_j(t) = \mathbb{P}(Z_j(t) = 1 \mid \mathcal{F}_0) \tag{103}$$

be its active-burning marginal probability. Throughout the timing bridge, t is numeric seconds relative to the forecast cutoff. The indicator activates the model's prescribed uniform source footprint over the entire cell. This is a spatial idealization: partial active coverage must be represented through the conditional whole-cell source density, rather than interpreted as a measured flame filling the raster cell. A cumulative burned-state indicator and a probability of first burning sometime during the next hour are different objects from $Z_j(t)$ and $p_j(t)$.

The combustion inputs for cell j are dry fuel loading $w_j \geq 0$ in $\mathrm{kg\,m^{-2}}$, phase-specific consumed fraction $f_j \in [0, 1]$, effective released heat per fuel mass $H_{c,j} > 0$ in $\mathrm{J\,kg^{-1}}$, and constant-emission duration $T_j > 0$ in seconds. Loading is mass per entire horizontal ground-cell area. The consumed fraction describes fuel consumed in the modeled phase, and effective released heat is a supplied value; neither is automatically corrected for moisture. Define the conditional released energy density and conditional heat-release-rate density by

$$e_j = w_j f_j H_{c,j} \qquad [\mathrm{J\,m^{-2}}], \tag{104}$$

$$Q_j = \frac{e_j}{T_j} = \frac{w_j f_j H_{c,j}}{T_j} \qquad [\mathrm{W\,m^{-2}}]. \tag{105}$$

For an active cell, the modeled total heat-release rate is $Q_j A_{\text{cell}}$ watts. With prescribed radiative fraction $\chi_j \in [0,1]$, the active-cell radiant rate is $\chi_j Q_j A_{\text{cell}}$ watts. The radiative fraction partitions released power; it is not receiver absorptivity or atmospheric transmissivity. The `CombustionParameters` object evaluates (105). Its fields can broadcast during construction, but the radiation model ultimately requires Q_j and χ_j to be either scalars or spatial maps with exactly the probability map's $[H, W]$ shape.

14.2 From a first-burning probability to active occupancy

The maintained checkpoint predicts first new cumulative burning during an exact one-hour interval, whose duration is $D = 3600$ s. Let N_j be that event, and define its *unconditional with respect to initial activity status* forecast probability, conditional on the supplied information $\mathcal{F}_0$, by

$$q_j = \mathbb{P}(N_j \mid \mathcal{F}_0). \tag{106}$$

Let U_j be the arrival time, relative to the cutoff, conditional on N_j. The temporal bridge adds the assumptions

$$U_j \mid (N_j, \mathcal{F}_0) \sim \text{Uniform}(0, D), \qquad Z_j^{\text{new}}(t) = \mathbf{1}_{N_j} \mathbf{1}_{\{U_j \le t < U_j + T_j\}}. \tag{107}$$

The conditional law is required only when $q_j > 0$. When $q_j = 0$, new activity is identically zero. Extend U_j arbitrarily off N_j, where its value cannot affect the indicator. A product of q_j and a conditional expectation given N_j is interpreted as zero in the $q_j = 0$ branch. The assigned model or scenario probability q_j is not an independently established true physical arrival probability. All expectation identities below are exact under the assigned probability law and supplied parameters. The trained predictor supplies q_j; it does not supply the uniform conditional arrival law or T_j. No assumption about dependence between arrival times in different cells is required for the expectations below.

For $0 \le t \le D$, the conditional interval of arrivals that remain active at t is $(\max(0, t - T_j), t]$. Its length is $\min(t, T_j)$, except for endpoints of zero probability under the continuous uniform law. Consequently,

$$p_j^{\text{new}}(t) = \mathbb{P}(Z_j^{\text{new}}(t) = 1 \mid \mathcal{F}_0) = q_j \frac{\min(t, T_j)}{D}. \tag{108}$$

Define the time-mean active probability and endpoint active probability by

$$\bar{p}_j^{\text{new}} = \frac{1}{D} \int_0^D p_j^{\text{new}}(t)\, dt, \qquad p_{j,D}^{\text{new}} = p_j^{\text{new}}(D). \tag{109}$$

If $T_j \le D$, split the integral at T_j:

$$\bar{p}_j^{\text{new}} = \frac{q_j}{D^2} \left(\int_0^{T_j} t\, dt + \int_{T_j}^D T_j\, dt \right) \tag{110}$$

$$= q_j \left(\frac{T_j}{D} - \frac{T_j^2}{2D^2} \right). \tag{111}$$

If $T_j \ge D$, the integrand is $q_j t / D$ over the whole forecast interval, so $\bar{p}_j^{\text{new}} = q_j/2$. Writing $\rho_j = \min(T_j, D)/D \in (0, 1]$ gives both cases in one expression:

$$\boxed{\bar{p}_j^{\text{new}} = q_j(\rho_j - \rho_j^2/2), \qquad p_{j,D}^{\text{new}} = q_j \rho_j.} \tag{112}$$

Equivalently, a cell arriving at time u burns for $\min(T_j, D - u)$ seconds within $[0, D]$, and $\bar{p}_j^{\text{new}} = q_j \mathbb{E}[\min(T_j, D - U_j) \mid N_j, \mathcal{F}_0]/D$. The corresponding expected radiant energy emitted by a newly igniting cell *within the forecast interval*, before atmospheric loss, is

$$A_{\text{cell}} \chi_j Q_j D \bar{p}_j^{\text{new}} = A_{\text{cell}} \chi_j e_j q_j \begin{cases} 1 - T_j/(2D), & T_j \le D, \\ D/(2T_j), & T_j \ge D. \end{cases} \tag{113}$$

This generally differs from $A_{\text{cell}}\chi_j e_j q_j$, because a cell arriving late can continue burning beyond the forecast interval. Thus the time-mean occupancy is the expected fraction of the forecast interval spent active; it is not an instantaneous state at the midpoint. For example, $T_j = 300$ s and $D = 3600$ s give $\rho_j = 1/12$, $\bar{p}_j^{\text{new}} = (23/288)q_j$, and $p_{j,D}^{\text{new}} = q_j/12$.

The probability q_j alone does not determine active occupancy. Arrivals concentrated near zero and arrivals concentrated near D can have the same q_j and different interval dose and endpoint activity. Equation (112) is valid for the implemented additional arrival scenario. The generic temporal utilities allow any positive numeric D, while `WildfireHeatModel` enforces $D = 3600$ because its maintained predictors were fitted on exact one-hour pairs.

14.3 Explicit contribution from fire already active at the cutoff

Cumulative burning by time zero does not establish current combustion. Existing fire is included only when the caller supplies both its initial active probability $p_{j,0}^{\text{old}} \in [0, 1]$ and its deterministic remaining active duration $R_j \geq 0$ seconds conditional on that initial state. The modeled old-fire contribution has active interval $[0, R_j)$:

$$p_j^{\text{old}}(t) = p_{j,0}^{\text{old}} \mathbf{1}_{\{0 \leq t < R_j\}}, \qquad \bar{p}_j^{\text{old}} = p_{j,0}^{\text{old}} \frac{\min(R_j, D)}{D}, \qquad p_{j,D}^{\text{old}} = p_{j,0}^{\text{old}} \mathbf{1}_{\{R_j > D\}}. \tag{114}$$

The strict endpoint inequality matters: $R_j = D$ gives extinction exactly at the endpoint and therefore zero endpoint activity. The API permits $R_j = 0$, which contributes zero throughout the modeled interval. The same prescribed conditional source density $\chi_j Q_j$ is applied to present fire as to new fire. The numerical checks require $R_j \geq 0$ but do not require $R_j \leq T_j$ or verify the remaining fuel inventory; consistency of an old-fire phase and its remaining duration is a caller-supplied physical assumption.

The pipeline requires positive initial active probability only in explicitly known already-burned cells. It sets the first-new-burning probability to zero in those cells and marks that zero valid even if the learned neighborhood is unsupported. Initial-active sources and possible first-new-burning sources therefore have disjoint cell support in this implementation. Then

$$\bar{p}_j = \bar{p}_j^{\text{new}} + \bar{p}_j^{\text{old}}, \qquad p_{j,D} = p_{j,D}^{\text{new}} + p_{j,D}^{\text{old}}. \tag{115}$$

These additions use unconditional event probabilities. More general same-cell combinations would require disjoint events and consistent conditioning; a probability conditional on initially being unburned would first need multiplication by the probability of that conditioning event. The implementation rejects overlapping positive initial activity and possible new burning instead of silently treating them as independent. If initial inputs are absent, $p_j^{\text{old}}(t)$ is set to zero as an explicit *new-ignition-only scope*; no inference of old-fire age is made.

14.4 Radiation geometry and the exact finite-cell transfer

All source cells are represented by horizontal emitting sheets at common height z_s, and all receivers lie at common height z_r in the same datum. The required separation is $\ell = z_s - z_r > 0$. The receivers are upward-facing horizontal surfaces with unit normal $\mathbf{n} = (0, 0, 1)$. The common datum can be shifted, even to negative heights; only ℓ affects this model. Terrain elevation, slope, and wind may affect the learned probability model, but they do not tilt a receiver, bend a flame, or change the sheet heights in the radiation operator.

For a horizontal receiver point $\mathbf{x}_i$ and horizontal source location $\mathbf{y} \in S_j$, where S_j is cell j's footprint, define

$$r(\mathbf{y}, \mathbf{x}_i) = \sqrt{\|\mathbf{y} - \mathbf{x}_i\|^2 + \ell^2}, \qquad \cos\theta = \frac{\ell}{r(\mathbf{y}, \mathbf{x}_i)}. \tag{116}$$

The angle θ is between the upward receiver normal and the direction from the receiver toward the source. An infinitesimal horizontal source area dA has isotropic radiant power density

$\chi_j Q_j$, giving conditional radiant rate $\chi_j Q_j \, dA$. Isotropic emission spreads that rate over $4\pi r^2$, and projecting it onto the receiver contributes the factor $\cos\theta$. For fixed scalar atmospheric transmissivity $\mathcal{T} \in [0,1]$, the resulting transfer is

$$K_{ij} = \frac{1}{4\pi} \int_{S_j} \frac{\ell}{(\|\mathbf{y} - \mathbf{x}_i\|^2 + \ell^2)^{3/2}} \, dA = \frac{\Omega_{ij}}{4\pi}. \tag{117}$$

Here Ω_{ij} is the solid angle in steradians subtended by the horizontal parallelogram at the receiver. The integral is dimensionless: ℓ/r^3 has unit m^{-2} and dA has unit m^2. The equality to solid angle uses the horizontal source sheet and parallel horizontal receiver normal. There is no additional Lambertian source-normal factor: the infinitesimal emitters are isotropic, rather than a solid-flame surface with a directional emission law. Arbitrary receiver normals or vertical receiving walls are not supported by the current API.[5]

For completeness, the implemented solid-angle calculation is as follows. Write the four source-corner vectors from the receiver as

$$\mathbf{v}_k = (\mathbf{x}_j - \mathbf{x}_i + \tfrac{1}{2}(\alpha_k \mathbf{u} + \beta_k \mathbf{v}),\ \ell), \quad (\alpha_k, \beta_k) \in \{(-1,-1),(1,-1),(1,1),(-1,1)\}, \tag{118}$$

in the displayed order $k = 0, 1, 2, 3$. For a triangle with corner indices (a_1, a_2, a_3), put $n_k = \|\mathbf{v}_k\|$ and define

$$D_{a_1 a_2 a_3} = n_{a_1} n_{a_2} n_{a_3} + (\mathbf{v}_{a_1} \cdot \mathbf{v}_{a_2}) n_{a_3} + (\mathbf{v}_{a_2} \cdot \mathbf{v}_{a_3}) n_{a_1} + (\mathbf{v}_{a_3} \cdot \mathbf{v}_{a_1}) n_{a_2}, \tag{119}$$

$$\Omega_{a_1 a_2 a_3} = 2 \operatorname{atan2}(\ell A_{\mathrm{cell}}, D_{a_1 a_2 a_3}). \tag{120}$$

The magnitude of the triple product for either triangular half of the cell is ℓA_{cell}. Splitting along diagonal $(0, 2)$ gives

$$\Omega_{ij} = \Omega_{012} + \Omega_{023}, \qquad K_{ij} = \frac{\Omega_{012} + \Omega_{023}}{4\pi}. \tag{121}$$

The routine constrains roundoff to $0 \le K_{ij} \le 1/2$. The use of finite cells and strictly positive separation replaces the singular same-location point-source approximation with a finite integral. There is no tunable minimum distance, added squared-distance constant, distance cutoff, or normalized spatial kernel. At a receiver under the interior of a cell, $\ell \downarrow 0$ gives $\Omega_{ij} \to 2\pi$ and $K_{ij} \to 1/2$.

In the far field, when a cell is small compared with its distance r_{ij} from the receiver, the transfer approaches

$$K_{ij} \simeq \frac{A_{\mathrm{cell}} \ell}{4\pi r_{ij}^3}, \qquad F_{ij} \simeq \frac{\mathcal{T} P_j^{\mathrm{rad}}}{4\pi r_{ij}^2} \frac{\ell}{r_{ij}}, \tag{122}$$

where $P_j^{\mathrm{rad}} = \chi_j Q_j A_{\mathrm{cell}}$ is active-cell radiant power. The factor ℓ/r_{ij} is required by the upward receiver orientation. As a separate consistency check, the infinite horizontal receiving plane intercepts $\mathcal{T}/2$ of total isotropic source power; a uniform infinite source sheet of density s gives flux $\mathcal{T}s/2$. Neither identity permits identifying the sum of finite-grid receiver samples with emitted source power.

Atmospheric transmissivity is a single supplied path-independent scalar. The implementation multiplies the final irradiance by $\mathcal{T}$, without changing the emitted radiant source power. It does not compute path length, humidity or smoke absorption, occlusion, shielding, or source temperature.

14.5 Why marginal probabilities suffice for the expected flux

Condition on $\mathcal{F}_0$ and on the supplied fixed physical parameters and geometry. The instantaneous incident radiant flux is the modeled random sum

$$F_i(t) = \mathcal{T} \sum_{j \in \mathcal{J}} K_{ij} \chi_j Q_j Z_j(t) \qquad [\mathrm{W\,m^{-2}}]. \tag{123}$$

Because the source domain is finite, expectation can be taken term by term:

$$\mathbb{E}[F_i(t) \mid \mathcal{F}_0] = \mathcal{T} \sum_{j \in \mathcal{J}} K_{ij} \chi_j Q_j \mathbb{E}[Z_j(t) \mid \mathcal{F}_0] \tag{124}$$

$$= \boxed{\mathcal{T} \sum_{j \in \mathcal{J}} K_{ij} \chi_j Q_j p_j(t)}. \tag{125}$$

This calculation requires no independence between the $Z_j(t)$. For example, perfectly correlated burning in two cells and mutually exclusive burning in those cells can yield identical marginal probabilities and thus identical expected fluxes, even though their fluctuation and exceedance probabilities differ. The present API computes an expectation, not those joint-distribution-dependent quantities.

The source strengths are prescribed conditional strengths, not random unconditional averages. If actual active source density Ξ_j in $\mathrm{W\,m^{-2}}$ were random, then in general

$$\mathbb{E}[Z_j \Xi_j \mid \mathcal{F}_0] = p_j \mathbb{E}[\Xi_j \mid Z_j = 1, \mathcal{F}_0], \tag{126}$$

whenever the conditional expectation is defined. It need not equal $p_j \mathbb{E}[\Xi_j \mid \mathcal{F}_0]$. Treating $\chi_j Q_j$ as fixed avoids silently imposing that incorrect factorization. Uncertainty in fuel, consumption, duration, radiative fraction, geometry, and transmission is not marginalized by this package.

The expected radiant source power returned by the radiation-only model is

$$\mathbb{E}[P_{\mathrm{radiant}}(t) \mid \mathcal{F}_0] = A_{\mathrm{cell}} \sum_{j \in \mathcal{J}} \chi_j Q_j p_j(t) \qquad [\mathrm{W}]. \tag{127}$$

This emitted source power excludes atmospheric loss. It is distinct from incident receiver flux or power.

Since the operator and source strengths are time independent, integration in time commutes with the finite sum. The interval-mean expected flux, expected dose per receiver area, and endpoint expected flux are respectively

$$\bar{F}_i = \mathcal{T} \sum_j K_{ij} \chi_j Q_j \bar{p}_j \qquad [\mathrm{W\,m^{-2}}], \tag{128}$$

$$\mathcal{E}_i = \int_0^D \mathbb{E}[F_i(t) \mid \mathcal{F}_0]\, dt = D \bar{F}_i \qquad [\mathrm{J\,m^{-2}}], \tag{129}$$

$$F_{i,D} := \mathbb{E}[F_i(D) \mid \mathcal{F}_0] = \mathcal{T} \sum_j K_{ij} \chi_j Q_j p_{j,D} \qquad [\mathrm{W\,m^{-2}}]. \tag{130}$$

Thus multiplying the exact interval mean by D is valid even when instantaneous occupancy changes during the hour. Multiplying endpoint flux by D would require a different constant-flux assumption and is not this pipeline's dose calculation.

14.6 Evaluation at raster centres or arbitrary points

For an aligned affine source and receiver grid with common heights, K_{ij} depends only on row and column displacement. The source matrix is $s_j = p_j \chi_j Q_j$. The implementation builds the transfer array with shape $[2H - 1, 2W - 1]$ and evaluates the linear convolution of that array with s. It pads both arrays to full convolution shape $[3H - 2, 3W - 2]$ before transforming them, multiplies their Fourier transforms, inverts the transform, and extracts rows $H - 1$ through $2H - 2$ and columns $W - 1$ through $2W - 2$. The Fourier transform accelerates this sum; it does not change the radiation law or approximate the finite-cell solid angles. Padding prevents a source near one edge from wrapping periodically to the opposite edge. Source values outside the supplied raster are excluded by a finite-domain assumption; this padding does not establish that unavailable source data are physically zero. Small negative floating-point values after the inverse transform

are replaced by zero because the exact mathematical sum is nonnegative. Input probabilities are validated rather than clipped.

Alternatively, `predict_at_points` accepts an $[N, 2]$ array of arbitrary horizontal projected receiver coordinates and sums the same exact cell transfers directly. Its default target and source chunk sizes are 128 and 1024, respectively. Chunking bounds temporary memory usage and has no physical meaning. Zero-strength cells can be skipped in this direct sum. Every receiver still lies on the configured common horizontal plane.

The raster output contains flux values at receiver cell *centres*, not averages of flux over receiver cells. If a small receiving area $A_i^{\mathrm{recv}} \geq 0$ is assumed to experience uniform flux, then its incident radiant power is $\bar{F}_i A_i^{\mathrm{recv}}$ watts, and its interval incident energy is $D\bar{F}_i A_i^{\mathrm{recv}}$ joules. A large receiving patch S requires the area integral $\int_S \bar{F}(\mathbf{x})\, dA$, or a sufficiently resolved numerical approximation. Multiplication of one point sample by the whole source-cell area is not generally that integral. None of these quantities is absorbed heat; that would additionally require an appropriate receiver absorption model.

For a separate sampled flux time series at strictly increasing seconds $t_0 < \cdots < t_M$, `integrate_heat_flux` uses the trapezoidal rule

$$\mathcal{E}_i^{\mathrm{trap}} = \sum_{m=0}^{M-1} \frac{F_i(t_m) + F_i(t_{m+1})}{2}(t_{m+1} - t_m). \tag{131}$$

This assumes linear interpolation between supplied samples. It does not reconstruct missing intervals or rapid unsampled variations. For piecewise-constant interval forecasts, the corresponding calculation is $\sum_m \bar{F}_{i,m}\Delta t_m$.

14.7 Unknown sources, explicit estimates, and monotone bounds

The radiation-only model rejects missing, masked, nonfinite, negative, or above-one active probabilities. The integrated pipeline can preserve unsupported learned probabilities using an explicit Boolean validity mask. Let $V \subseteq \mathcal{J}$ be the set of resolved q_j, after setting known already-burned first-event probabilities to zero. A valid entry must be finite and in $[0, 1]$. An invalid entry is retained as NaN even when its raw stored number is finite; validity is part of the data contract.

An optional externally justified probability map can be supplied for unsupported entries. It changes only those entries and is recorded using an externally-supplied-probability mask. The pipeline also retains the original validity mask, so a supplied estimate cannot be mistaken for the checkpoint's supported prediction.

For entries still unresolved, let

$$q_j^- = \begin{cases} q_j, & j \in V, \\ 0, & j \notin V, \end{cases} \qquad q_j^+ = \begin{cases} q_j, & j \in V, \\ 1, & j \notin V. \end{cases} \tag{132}$$

Put $g_j = \rho_j - \rho_j^2/2$ and $b_j = \bar{p}_j^{\mathrm{old}}$. For any complete map q^* consistent with the known entries and $0 \leq q_j^* \leq 1$, define

$$\bar{F}_i(q^*) = \mathcal{T} \sum_j K_{ij}\chi_j Q_j(g_j q_j^* + b_j). \tag{133}$$

Every coefficient in this sum is nonnegative. Therefore the implemented interval-mean bounds satisfy

$$\boxed{\bar{F}_i^- := \bar{F}_i(q^-) \;\leq\; \bar{F}_i(q^*) \;\leq\; \bar{F}_i(q^+) =: \bar{F}_i^+.} \tag{134}$$

Endpoint bounds replace g_j by ρ_j and b_j by $p_{j,D}^{\mathrm{old}}$. Dose bounds are $D\bar{F}_i^-$ and $D\bar{F}_i^+$. The lower and upper substitutions are mathematical uncertainty endpoints, not observations or default

assumptions about missing fire. The finite known-source contribution is $\bar{F}_i^-$, including any explicitly supplied present-active sources. It is not the unknown full expectation.

With $\ell > 0$ every nonempty source cell has positive geometric transfer to every finite receiver point. Consequently, an unresolved q_j for which $\mathcal{T}\chi_j Q_j > 0$ can affect the entire receiver field, not merely its corresponding pixel. The pipeline marks the full expected mean flux and expected dose as NaN throughout the receiver raster if any such emitting source remains unresolved, and sets `complete=False`. The bounds and known-source contribution remain finite. If every unresolved cell has zero source density, zero radiative fraction, or $\mathcal{T} = 0$, the full flux is nevertheless determined, so `complete=True` is possible even with unresolved probability entries. The occupancy entries themselves remain unknown.

These are bounds for fixed supplied combustion, timing, transmission, geometry, and finite source domain. They do not bound unmodeled transport, outside-domain fires, parameter error, active flames omitted from the present-fire scope, or errors in the learned probability model.

14.8 Surface-flame scenarios and position selection

The point-residence construction and its reference-fuel assumptions are described in Section 13. A residence duration describes flaming at a representative fuel location, not time to cross the entire 375 m raster cell or time until all smoldering and trailing combustion ends.

For a supplied known or estimated arrival a_j seconds and query time t seconds sharing a common origin, `active_flames_from_arrival` defines the scenario indicator and remaining duration by

$$z_j^{\text{scenario}}(t) = \mathbf{1}_{\{a_j \le t < a_j + \tau_j^{\text{fl}}\}}, \tag{135}$$

$$R_j^{\text{scenario}}(t) = \begin{cases} \tau_j^{\text{fl}} - (t - a_j), & a_j \le t < a_j + \tau_j^{\text{fl}}, \\ 0, & \text{otherwise.} \end{cases} \tag{136}$$

If a reconstructed arrival $\widehat{a}_j$ is recorded in hours from the event origin and the query uses that same origin, its conversion is $a_j = 3600\widehat{a}_j$ seconds. If time zero instead denotes the forecast cutoff, subtract the cutoff elapsed time from that converted arrival before forming $t - a_j$. Mixing event-relative hours and cutoff-relative seconds would give an incorrect activity indicator. The implementation compares elapsed age $t - a_j$ with τ_j^{fl}, avoiding an unnecessary overflow from forming $a_j + \tau_j^{\text{fl}}$. It permits negative arrival and query times when their common origin warrants them. At exactly $a_j + \tau_j^{\text{fl}}$, the location is inactive. Invalid arrivals remain NaN and unclassified. An estimated reconstructed arrival yields a nominal scenario; this function does not integrate arrival uncertainty or identify measured flame positions.

Define $\rho_j^{\text{fl}} = \min(\tau_j^{\text{fl}}, D)/D$. The flame-specific mean is $q_j[\rho_j^{\text{fl}} - \frac{1}{2}(\rho_j^{\text{fl}})^2]$, and the endpoint is $q_j\rho_j^{\text{fl}}$. These specialize the timing bridge to surface-flame residence; they agree with the heat-emission bridge only when the caller explicitly sets $T_j = \tau_j^{\text{fl}}$. For uncertain new arrivals, `forecast_active_flames` returns q_j, the mean flame occupancy specialized above, and an endpoint map containing $q_j\rho_j^{\text{fl}}$. Existing flames are excluded from this forecast unless supplied separately through the integrated heat pipeline. Unknown probabilities remain NaN. A requested selection cutoff $\alpha \in (0, 1]$ returns exactly the valid cells whose instantaneous probability is at least α, together with their zero-based row and column indices and projected cell-centre coordinates. There is no implicit selection threshold. Thresholding is a display or candidate-selection rule; it does not establish detection accuracy or calibration against active-flame observations. Empty selections remain usable arrays of shape $[0, 2]$ for indices and coordinates and $[0]$ for probabilities.

14.9 The composed checkpoint-to-heat procedure

The public `WildfireHeatModel` executes the following sequence for one spatial snapshot.

1. Load the saved checkpoint, identify either the maintained direct environment-conditioned convolution or survival-intensity convolution architecture, and reuse its saved channel schema, categorical vocabulary, fitted weights, and training-only normalization statistics. The integration does not train or fit preprocessing. Checkpoint path, SHA-256, architecture, selected epoch, channel information, and device are recorded. CPU is the default device.

2. Establish a common projected affine grid, complete combustion scenario, radiative fraction, source/receiver heights, and transmission scalar. If a `FlameResidenceTime` object is also supplied, its duration must equal `combustion.burning_duration_s` exactly at every represented cell. The same duration then governs active occupancy and the denominator in conditional heat release. Consumed fraction must concern that modeled flaming phase.

3. Pass the current cumulative burned state, current environmental tensor, exact channel order, and separate validity masks to the saved predictor. Use a one-hour interval, `dt_hours=1.0`, and an unknown boundary, `boundary="unknown"`. Only a single $[H, W]$ snapshot, possibly with singleton leading dimensions, is accepted. For sample-based heat prediction, the affine coefficients must agree with the configured radiation grid to absolute tolerance 10^{-9} and zero relative tolerance.

4. Take the predictor's evaluated probability and validity outputs. For a direct convolution this means its bounded evaluation probability, not its unbounded signed raw score. Resolve first-new-burning probability to zero in known already-burned cells. Retain unsupported entries and separately record any caller-supplied estimates.

5. Convert new-event probabilities to mean and endpoint active occupancy using the uniform-arrival and constant-duration scenario. Add only explicitly supplied initial-active contributions with known already-burned support and remaining durations.

6. Apply the fixed radiation operator to lower and upper mean occupancies and lower and upper endpoint occupancies. Return a complete expected mean only when unknown emitting contributions are absent. Otherwise return the finite known-source contribution and monotone bounds, preserving NaN for the unresolved full expectation.

7. Multiply mean flux and its bounds by 3600 s to obtain interval radiant dose per receiver area. Preserve probability masks, externally supplied masks, temporal and physical assumptions, coverage counts, completeness, and provenance in the result.

The sample adapter does not consume arrival surfaces, arrival uncertainty, or next-hour labels to run the checkpoint. Nevertheless, a current burned state reconstructed retrospectively using an arrival surface and final extent remains a retrospective input. Restricting environmental intervals to end before the cutoff does not convert that reconstruction into an operationally observed current fire state.

14.10 Exact scope of the saved demonstrations and verification

The saved `integrated_example.py` demonstration uses the Fish Creek event (identifier `WY-BTF-002416_FishCreek`). Its cutoff is `2024-08-23T00:00:00Z`; it uses the signed direct 5×5 checkpoint, and a 103×73 grid in EPSG:5070 with 375 m cells. Its specified parameters are $H_{c,j} = 18{,}000{,}000\ \mathrm{J\,kg^{-1}}$, $T_j = 300$ s, $f_j = 0.10$, $\chi_j = 0.30$, $z_s = 5$ m, $z_r = 0$ m, and $\mathcal{T} = 1$. The ecological loading adapter uses an explicit $10\ \mathrm{kg\,m^{-2}}$ loading scenario for unmapped fractions. Those are illustrative scenario values, not fitted heat-flux parameters. The saved reference-versus-uniform comparison keeps the same probability map and compares ecological loading with constant $10\ \mathrm{kg\,m^{-2}}$ loading. It establishes sensitivity to that input, not an accuracy improvement. The historical summary records 225 resolved probabilities, 7294 unresolved probabilities, and `complete_expected_flux_available=False`; the finite plotted known-source contribution must therefore not be relabeled as full expected heat.[4]

The separate `active_flame_example.py` uses the same default event, cutoff, and checkpoint but constructs a spatial reference-fuel residence map, rather than using the heat demonstration's constant 300 s duration. Its default unsupported-code SAV fallback is explicitly 2000 ft^{-1}, and the default candidate cutoff is 0.001. The saved example preserves a zero nominal-current-flame selection and zero forecast candidates at that cutoff, while also exporting all positive endpoint probabilities. A zero selected count is retained as a result of the specified scenario and threshold. It is not evidence that no flames physically existed. The reconstructed-arrival uncertainty is much larger than the point-scale residence duration, and is not integrated into the nominal indicator.

The current source tests in `tests/heat/test_model.py`, `tests/heat/test_temporal.py`, `tests/heat/test_pipeline.py`, `tests/heat/test_flames.py`, and `tests/heat/test_flame_integration.py` specify independent mathematical checks: source-area Gauss–Legendre integration, analytic rectangle solid angles, the orientation-corrected far-field limit, half-power interception by an infinite receiver plane, equal expectations for distinct dependence structures with identical marginals, affine rotation/shear, lack of periodic wrap, temporal indicator quadrature, endpoint extinction conventions, unknown-source bounds, units, and preservation of saved preprocessing and masks. These checks verify the stated mathematical model and software contracts. Historical validation-report files record earlier runs, sometimes using paths predating test relocation. They are not substitutes for a fresh current test result and do not constitute measured-wildfire heat or active-flame calibration.

15 Step-by-step worked calculations and public interfaces

15.1 A complete one-cell calculation with an analytic answer

This example isolates the timing and radiation equations. It is synthetic and is not a fitted fire scenario. Consider one square 10 m cell, with its receiver at the horizontal center, source height 5 m, receiver height 0 m, and transmissivity $\mathcal{T} = 1$. Supply $q = 0.8$ for first new burning in the next $D = 3600$ seconds, no initial active fire, and

$$w = 1\ \mathrm{kg\,m^{-2}}, \quad f = 1, \quad H_c = 18{,}000{,}000\ \mathrm{J\,kg^{-1}}, \quad T = 120\ \mathrm{s}, \quad \chi = 0.3. \tag{137}$$

Each step is now fully determined:

1. The cell area is 100 m^2, but dry fuel loading is already mass per unit whole-cell ground area. Released energy density is $e = wfH_c = 18{,}000{,}000\ \mathrm{J\,m^{-2}}$.

2. Conditional heat-release density is $Q = e/T = 150{,}000\ \mathrm{W\,m^{-2}}$. An active cell therefore releases 15,000,000 W in total, of which 4,500,000 W is radiative under $\chi = 0.3$.

3. The capped duration ratio is $r = T/D = 1/30$. Under the uniform-arrival law, the mean and endpoint activity probabilities are

$$\overline{p} = 0.8\left(\frac{1}{30} - \frac{1}{1800}\right) = \frac{59}{2250} \simeq 0.02622222, \qquad p(D) = \frac{0.8}{30} = \frac{2}{75} \simeq 0.02666667. \tag{138}$$

4. For a centered rectangle with half-width $a = 5$ m, half-height $b = 5$ m, and vertical separation $\ell = 5$ m, its solid angle is

$$\Omega = 4\arctan\left(\frac{ab}{\ell\sqrt{\ell^2 + a^2 + b^2}}\right) = 4\arctan(1/\sqrt{3}) = \frac{2\pi}{3}. \tag{139}$$

   Thus the finite-cell transfer $\Omega/(4\pi)$ is 1/6.

5. Mean incident radiant flux is

$$\overline{F} = \mathcal{T}\,\overline{p}\,\chi Q\frac{1}{6} = \frac{590}{3}\ \mathrm{W\,m^{-2}} \simeq 196.6666667\ \mathrm{W\,m^{-2}}. \tag{140}$$

Endpoint flux is $F_D = p(D)\chi Q/6 = 200\ \mathrm{W\,m^{-2}}$. Neither equals the next-hour transition probability multiplied directly by the conditional active-cell flux.

6. Expected interval incident energy is

$$\mathcal{E} = D\overline{F} = 708{,}000\ \mathrm{J\,m^{-2}}. \tag{141}$$

This is a point receiver's dose per receiver area, not total energy received by the whole ground cell.

7. All source probabilities and physical parameters are supplied and finite, so this single-source calculation is complete. A missing q would instead produce a range by substituting 0 and 1, under the same fixed scenario.

The public objects evaluate this example without any wildfire checkpoint:

```
import numpy as np
from expected_heat_flux import (
    AffineGrid, CombustionParameters, ExpectedHeatFluxModel,
    ignition_to_active_probability,
)

combustion = CombustionParameters(
    fuel_load_kg_m2=1.0,
    heat_of_combustion_j_kg=18_000_000.0,
    burning_duration_s=120.0,
    consumed_fraction=1.0,
)
timing = ignition_to_active_probability(
    np.array([[0.8]]), 120.0, 3600.0,
)
radiation = ExpectedHeatFluxModel(
    grid=AffineGrid.from_cell_size(10.0),
    heat_release_rate_density_w_m2=(
        combustion.heat_release_rate_density_w_m2
    ),
    radiative_fraction=0.3,
    emission_height_m=5.0,
    receiver_height_m=0.0,
    atmospheric_transmissivity=1.0,
)
mean = radiation.predict(timing.mean_active_probability)
endpoint = radiation.predict(timing.endpoint_active_probability)
mean_flux = mean.incident_heat_flux_w_m2[0, 0]
endpoint_flux = endpoint.incident_heat_flux_w_m2[0, 0]
np.testing.assert_allclose(mean_flux, 590.0 / 3.0)
np.testing.assert_allclose(endpoint_flux, 200.0)
np.testing.assert_allclose(mean_flux * 3600.0, 708_000.0)
```

The code names, call signatures, and assertions above were executed against the current package during document preparation. Only the array's active probability is passed to the independent radiation object; no event-transition probability is implicitly reinterpreted inside that object.

15.2 A residence-based endpoint example

With characteristic SAV 2000 $\mathrm{ft^{-1}}$, the correctly converted SI value is $2000/0.3048\ \mathrm{m^{-1}}$. Anderson–Andrews residence is $\tau^{\mathrm{fl}} = 11.52$ s. For $q = 0.8$ and the same one-hour horizon,

$$p^{\mathrm{fl}}(D) = 0.8\frac{11.52}{3600} = 0.00256, \qquad \overline{p}^{\mathrm{fl}} = 0.8\left(0.0032 - \frac{0.0032^2}{2}\right) = 0.002555904. \tag{142}$$

The calculation does not reduce the probability of burning sometime during the hour; that remains 0.8. It changes the event whose probability is being asked for. Endpoint flame activity requires ignition sufficiently close to the endpoint to remain active under the supplied short residence duration.

15.3 Loading exactly the saved model's input contract

Run the following from the directory containing the root `README.md` and `pyproject.toml`, using the project's Python 3.12 environment. Its dependency pins include NumPy 1.26.4 and PyTorch 2.5.1. The source/data paths below are repository-relative for portability. An installed code-only wheel requires explicit access to the separately retained data and checkpoint.

```
from pathlib import Path
from wildfire_burned_area_hourly import load_model

root = Path.cwd()
checkpoint = (
    root / "wildfire_burned_area_hourly/probability_model"
    / "training_runs/20261007_1h_direct_convolution_signed"
    / "5x5/fitted/checkpoint.pt"
)
model = load_model(checkpoint, device="cpu")
with model.load_data(
    root / "wildfire_burned_area_hourly/environmental_data",
    events=["WY-BTF-002416_FishCreek"],
    sequence_length=1,
    time_range=("2024-08-23T00:00:00Z", "2024-08-23T00:00:00Z"),
) as dataset:
    assert len(dataset) == 1
    sample = dataset[0]

prediction = model.predict_sample(sample)
assert tuple(sample["channel_names"]) == model.channel_names
assert tuple(prediction.probability.shape) == (1, 1, 103, 73)
print(int(prediction.valid.sum()))  # 225 in the verified example
```

The example's 7519 raster locations comprise 225 valid probabilities and 7294 unknown probabilities. The CPU run during authorship produced maximum known probability 0.09081834554672241. These counts describe prediction coverage, not accuracy; the example itself does not consume future labels for inference. The checkpoint predicts raw and clipped values separately. Use `prediction.probability` and `prediction.valid` for probability calculations, and retain `prediction.raw_score` for raw-score analysis.

To connect this snapshot to heat, construct a metric `AffineGrid` from the sample's six affine coefficients, supply complete combustion inputs and geometry, and call `model.heat_model(...).predict_sample(sample)`. The result's `complete` field determines whether the full supplied-domain expected heat is known. Export arrays, their masks, affine transform, CRS, timestamp, physical assumptions, and provenance together. A local valid probability does not suffice to establish a locally complete heat prediction when radiation from another source is unknown.

16 What the calculation establishes, and what remains unspecified

16.1 Learned quantities, supplied quantities, and absent quantities

Stage	Computed or learned here	Additional information not supplied by that stage
Arrival product	Retrospective location-specific time and kriging standard deviation	Live arrival measurements, calibrated true-arrival distribution, or support beyond final perimeter.
Environmental alignment	Common grid values, categories, support metadata, and masks	Finer independent information than each native source or proven cutoff publication availability.
Probability predictor	Conditional next-hour transition estimates on its saved input and contact contract	Within-hour arrival density, an active flame front, spontaneous ignition, or physical propagation trajectories.
Residence assignment	Deterministic fuel-reference or explicit duration scenario	Measured flame geometry, a random duration law, crown/residual duration, or species-wide validated duration.
Occupancy bridge	Activity marginals under uniform arrival and constant duration	Learned arrival-time shape, existing-fire age, or a joint distribution of activity across cells.
Radiation operator	Incident mean flux, endpoint flux, dose, and conditional missing-source bounds	Heat variance, exceedance probability, absorbed flux, temperature, convection, terrain shielding, or exterior-source contribution.

16.2 Why expectation cannot determine a danger probability

Let receiver flux be $F(x) = \sum_i k_i(x) Z_i$, with deterministic nonnegative source-transfer coefficients $k_i(x)$. Marginal active probabilities determine $\mathbb{E}[F(x)] = \sum_i k_i(x) p_i$. They do not determine

$$\operatorname{Var}(F(x)) = \sum_i k_i(x)^2 p_i(1-p_i) + 2 \sum_{i<j} k_i(x) k_j(x) \operatorname{Cov}(Z_i, Z_j). \tag{143}$$

Covariances and the full joint law are absent. For a concrete counterexample, two sources of unit strength can each have marginal probability 0.4. If they are perfectly correlated, total flux is 2 with probability 0.4 and 0 with probability 0.6. If they are mutually exclusive, total flux is 1 with probability 0.8 and 0 with probability 0.2. Both have mean 0.8, but $\mathbb{P}(F > 1.5)$ is 0.4 in the first case and zero in the second. Thus a threshold applied to the expected field is a threshold on a mean scenario output, not an estimated probability of harmful heat exposure.

16.3 Conditions needed for an operational interpretation

An operational evaluation would need a current fire-state product actually available at each cutoff, feature availability and latency records, future outcomes independent of reconstruction inputs, sufficient independent events, and appropriate calibration of the intended endpoint. Active flames would add event-specific arrival-age and duration evidence at a commensurate time scale. Heat exposure would add fuel/consumption/radiation measurements and validation for the intended receiver geometry and heat mechanism. None of these missing quantities is inferred merely from a passing numerical test, stored checkpoint, map image, or the algebraic timing bridge.

A Source map and evidence conventions

Paths in this appendix are relative to the project root. The companion `documents/wildfire_methods_source_audit.json` records SHA-256 hashes of the inspected implementation and key saved records at document creation. It also records current verification separately from saved historical reports. The document is a description of that snapshot; later code edits require checking these bindings before treating it as unchanged documentation.

Method / evidence	Primary local location
Published reconstruction	`wildfire_burned_area_hourly/sources/zenodo_21788344/extracted/code/` and its retained `README.md`, `results/`, and original rasters.
Source provenance	`wildfire_burned_area_hourly/source_manifest.json` and `records/`.
Environmental acquisition	`wildfire_burned_area_hourly/environmental_data/acquire_environment.py`, `enrich_environment.py`, `audit_source_quality.py`, and `acquire_biome.py`.
Tensor and label contract	`wildfire_burned_area_hourly/pytorch_loader/dataset.py`.
Preparation and sampling	`wildfire_burned_area_hourly/probability_model/prepare_training.py` and `filter_contact.py`.
Exponential family	`wildfire_burned_area_hourly/probability_model/model.py`, `train.py`, and `inference.py`.
Signed direct family	`wildfire_burned_area_hourly/probability_model/direct_convolution/model.py`, `train.py`, `inference.py`, and `evaluate.py`.
Schema and scoring	`wildfire_burned_area_hourly/probability_model/input_schema.py`, `metrics.py`, and `broader_comparison.py`.
Saved kernel comparison	`wildfire_burned_area_hourly/probability_model/training_runs/20261007_1h_direct_convolution_signed/` and `20261007_1h_kernel_sweep/`.
Public model API	`wildfire_burned_area_hourly/api.py` and `api_example.py`.
Ecological loading	`expected_heat_flux/ecology.py`.
Residence scenarios	`expected_heat_flux/residence.py`, `sources.json`, and `documents/flame_residence_literature.md`.
Occupancy and flames	`expected_heat_flux/temporal.py` and `flames.py`.
Radiation and composition	`expected_heat_flux/model.py` and `pipeline.py`.
Saved heat demonstrations	`expected_heat_flux/outputs/integrated_demo/summary.json` and `active_flame_demo/summary.json`.
Current contract tests	`tests/loader/`, `tests/probability/`, `tests/heat/`, and `tests/test_api.py`.

Claims labeled "saved" describe the preserved experiment or demonstration, including its configured parameters and date. Statements about current equations and contracts were checked against the implementation during authorship. No probability models were refitted and no publisher study was rerun for this document. Source-data integrity validation is distinct from predictive or physical validation. In particular, the recorded full environmental audit and its later biome extension explicitly distinguish previously checked unchanged files from newly reread raster bands.

B Reproducibility and numerical checks

The document itself is a standalone LaTeX source: all prose, mathematics, tables, examples, and bibliography are contained in one `.tex` file. It requires ordinary TeX packages, but no external figure, bibliography, data, or code file to typeset. The companion PDF is an exported build of that source. LaTeX compilation, resolution of cross-references/citations, final-log diagnostics, and rendered-page layout are checked before delivery.

The one-cell calculation in Section 15 is checked through the public temporal and radiation interfaces. The saved CPU inference example is rerun to verify shape, schema, counts, and a known-probability range. Selected residence and ecological-reader contract tests are run against the current project source. Exact commands, counts, outputs, and source hashes are recorded in the companion audit. These checks support the documented mathematical and software behavior; they are not a new performance or field-calibration study.

References

[1] C. Choi, H. Seo, and J. Cha. *Wildfire time-of-arrival dataset and method code: variance-weighted kriging of satellite active-fire detections, evaluated in the United States and applied to South Korea*. Zenodo, record 21788344, 26 September 2026. `https://doi.org/10.5281/zenodo.21788344`. Companion article: *Remote Sensing* 18(19), 3321 (2026), `https://doi.org/10.3390/rs18193321`. Record metadata and archived method code were inspected locally.

[2] Choi, Seo, and Cha, retained dataset method code. `preprocess.py`, `toa_kriging.py`, `produce_rasters.py`, `block_cv.py`, and `evaluate_sc.py` under the archived code directory identified in Appendix A. Units and source conventions follow these archived files.

[3] WildfireGuardian probability implementation and saved experiment records, inspected 8 October 2026. Sources and records are located in `wildfire_burned_area_hourly/probability_model/`; see Appendix A and the companion source-hash audit.

[4] WildfireGuardian radiation, temporal, flame, and composed API implementation, inspected 8 October 2026. `expected_heat_flux/model.py`, `temporal.py`, `pipeline.py`, and `flames.py`; saved demonstrations are explicitly identified in the text.

[5] K. J. Overholt. *Verification and Validation of Commonly Used Empirical Correlations for Fire Scenarios*. NIST Special Publication 1169, March 2014, Section 6.1, equation (6.1), printed p. 39 (PDF page 53). `https://doi.org/10.6028/NIST.SP.1169`. `https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.1169.pdf`. The isotropic point-source law includes the receiver-orientation cosine. The local radiation method integrates that differential source law over a horizontal finite footprint; this is a package modeling extension. A. Angelidis, *Spherical Springs*, University of Otago Technical Report OUCS-2004-01, 2004, Section 3.2, printed p. 2, `https://cs.otago.ac.nz/research/publications/oucs-2004-01.pdf`, records the triangle solid-angle formula and attributes it to A. van Oosterom and J. Strackee, *The Solid Angle of a Plane Triangle*, IEEE Transactions on Biomedical Engineering BME-30(2), 125–126 (1983), `https://doi.org/10.1109/TBME.1983.325207`. The accessible NIST and Otago PDFs were checked during preparation; the original IEEE paper is an attribution from the accessible report, not a claimed full-text inspection. The complete reference record is retained in `expected_heat_flux/sources.json`.

[6] WildfireGuardian source and saved artifacts inspected on 8 October 2026. `expected_heat_flux/residence.py`, `expected_heat_flux/ecology.py`, `expected_heat_flux/sources.json`, `tests/heat/test_residence.py`, `tests/heat/test_ecology.py`,

documents/flame_residence_literature.md, and expected_heat_flux/outputs/active_flame_demo/summary.json. The separate 300 s integrated combustion scenario is recorded in expected_heat_flux/outputs/integrated_demo/summary.json.

[7] LANDFIRE. *Fuel Characteristic Classification System Fuelbeds*, official product description and loading-table link. https://www.landfire.gov/fuel/fccs. Product description inspected online on 8 October 2026. Reference loading table: https://www.landfire.gov/sites/default/files/CSV/LF_ConsumeLoadings.csv. The existing workflow uses archived inputs with recorded source epochs; the current web page is not substituted for the archived raster release.

[8] S. J. Prichard, D. V. Sandberg, R. D. Ottmar, E. Eberhardt, A. Andreu, P. Eagle, and K. Swedin. *Fuel Characteristic Classification System version 3.0: technical documentation*. USDA Forest Service, Pacific Northwest Research Station, General Technical Report PNW-GTR-887, 2013. https://doi.org/10.2737/PNW-GTR-887. Official agency record and abstract: https://research.fs.usda.gov/treesearch/45283.

[9] H. E. Anderson. *Heat transfer and fire spread*. USDA Forest Service, Research Paper INT-69, 1969, printed pp. 15–16. https://research.fs.usda.gov/download/treesearch/40489.pdf.

[10] P. L. Andrews. *The Rothermel surface fire spread model and associated developments: A comprehensive explanation*. USDA Forest Service, Rocky Mountain Research Station, General Technical Report RMRS-GTR-371, 2018. https://doi.org/10.2737/RMRS-GTR-371. Official PDF: https://research.fs.usda.gov/download/treesearch/55928.pdf. Surface-area weighting: Sections 3.2.3–3.2.5, printed pp. 21–24. Residence relation and scope: Section 4.2, printed p. 27. These sections were checked in the available official-PDF local snapshot; the browsing service could not retrieve the large PDF in this audit.

[11] J. H. Scott and R. E. Burgan. *Standard fire behavior fuel models: a comprehensive set for use with Rothermel's surface fire spread model*. USDA Forest Service, Rocky Mountain Research Station, General Technical Report RMRS-GTR-153, 2005. https://doi.org/10.2737/RMRS-GTR-153. https://research.fs.usda.gov/download/treesearch/9521.pdf. Component parameters: Table 7, printed p. 18. Common constants: p. 17. Dynamic curing: p. 6. The source GS4 summary SAV differs from its table-derived value; the implementation records and resolves this as explained in the text. The relevant local official-PDF snapshot was inspected.

[12] N. D. Burrows. *Experimental development of a fire management model for jarrah (Eucalyptus marginata Donn ex Sm.) forest*. PhD thesis, Australian National University, 1994. Government-hosted original thesis: https://library.dbca.wa.gov.au/FullTextFiles/016094.pdf. Individual-particle methods: printed p. 67 (PDF page 76). Implemented diameter equation: printed p. 69, equation 6-1 (PDF page 78). These scanned pages were visually checked in the available local PDF; the browsing service could not retrieve the large PDF in this audit.